



Faculty of Computing and Information Technology

AWS Cloud Challenge: Deploying a Scalable and Secure Booking & Ticketing Platform

DFT2 (GROUP 12)

Assignment

**AMIT3253 Cloud Computing for
Business (202605)**

Tutor: YAP XIANG LOO WINNIE

Date of Submission: 7/9/2026

Name (Block Capital)	Student ID
CHENG JUN YU	25WMD07989
KHOO SEE ZE	25WMD08818
LOW KAR WAI	25WMD07972

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Group Contract Form

Faculty: Faculty Of Computer Science

Programme: Diploma in Information Technology

Year of Study: Y2

Semester: S1

Course Title: CLOUD COMPUTING FOR BUSINESS

Course Code: AMIT3253

Group Assignment/ Project Title: Swimming pool booking system

Name of Tutor: YAP XIANG LOO WINNIE

Tutorial Group: G12

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date
Cheng Jun Yu 2507989	AWS Service: VPC、Subnets(public/private)、Internet Gateway、Route Tables、Security Groups、NAT Gateway、RDS, EC2 、AMI、Launch Template、Auto Scaling Group、Target Group、Application Load Balancer, S3, SSM Parameter Store Report part: Secure, High Performing	 7/9/2026
Khoo See Ze 2508818	AWS Service: SNS + Lambda Report part: Functional, AWS Architecture Design, Load Balanced	 7/9/2026
Low Kar Wai 2507972	AWS Service: CloudWatch (Alarm+Dashboard) , WAF Report part: Solution Overview Diagram, Cost Optimized, Scalability	 7/9/2026

Member Profiles

Name (Block Capital)	Contact number	Profile Photo
CHENG JUN YU	+6011-2800 3667	
KHOO SEE ZE	+6011-56507798	
LOW KAR WAI	+6014-9815779	

Assignment Rubrics

Part 1: Reporting and Documentation (70 marks)

Criteria	Unsatisfactory (1-3)	Needs Improvement (4-5)	Good (6-8)	Excellent (9-10)	Mark
AWS Architecture Diagram (5m) & Solution Overview Diagram (5m)	Diagram is missing, heavily incomplete, or uses non-standard cloud icons. Network boundaries (VPC, subnets) are omitted or incorrect. Flow diagram is missing, or the sequential flow of data from the end-user to the backend database is entirely absent or illogical.	A diagram is provided but lacks detail. Component relationships are vague, or structural layout (public vs. private subnets) is poorly defined. A diagram is provided, but the transactional flow is confusing, or the user journey lacks clear step-by-step logic.	The diagram is complete and mostly accurate. Clearly displays the custom VPC layout, subnets, EC2 instances, ELB, ASG, and RDS with minor labeling gaps. The diagram successfully maps out the functional user journey and data pathway from the web interface down to the data tier with minor logical gaps.	Exceptional, production-ready infrastructure diagram. Perfectly details the entire AWS environment, security group boundaries, and gateways. A highly detailed visual mapping of the functional data journey. Flawlessly illustrates how requests progress through the load balancer, application, and database transactions.	
Functional	The booking platform is non-functional. Users cannot view slots/events or make reservations, and database operations fail completely.	The website loads, but core booking components are broken. Database connections are unstable, or errors occur during data submission.	The booking platform is functional. Users can browse events and book tickets successfully, though minor UI/UX bugs or validation errors exist.	The application is fully operational and seamless. Users can flawlessly browse events, select quantities, complete bookings, and verify data updates in the database.	

Assignment Rubrics (cont.)

Part 1: Reporting and Documentation (70 marks)

Criteria	Unsatisfactory (1-3)	Needs Improvement (4-5)	Good (6-8)	Excellent (9-10)	Mark
Secure	Security is ignored. The database is exposed to the public internet, or Security Groups are wide open (0.0.0.0/0) across all ports.	A VPC is present, but resources are misplaced (e.g., database sits in a public subnet). Security groups are overly permissive.	Network segregation is mostly correct. The database sits in a private subnet, and Security Groups restrict basic traffic with minor rule optimization gaps.	Exemplary security implementation. Custom VPC has isolated public and private subnets. Both the EC2 application instances and the RDS database are strictly isolated within private subnets. Public access is restricted entirely to the Application Load Balancer, which demonstrates proper request handling by securely routing internet traffic to the private application tier. Security Groups act as tight, least-privilege firewalls.	
Cost Optimized	Cost report is missing or completely inaccurate. Overpriced or unnecessary enterprise-grade AWS resources are chosen without consideration of the lab budget.	Cost estimation is provided but lacks detail or calculation structure. Resource choices are poorly justified or slightly push the budget limits.	A complete AWS Pricing Calculator report is included with approximate estimations. Resource choices (EC2/RDS sizing) are reasonably justified for a POC.	Provides a highly comprehensive, exported AWS Pricing Calculator report. Resource sizing is expertly optimized for a budget-conscious POC with strong written justifications.	

Assignment Rubrics (cont.)

Part 1: Reporting and Documentation (70 marks)

Criteria	Unsatisfactory (1-3)	Needs Improvement (4-5)	Good (6-8)	Excellent (9-10)	Mark
High Performing	No stress or load testing was conducted. The application crashes immediately or becomes completely unresponsive under minimal traffic.	Load testing was attempted, but results are missing analysis or screenshots. The application experiences severe latency or errors under mild simulation.	Stress testing was successfully executed using a simulation tool. The system remains stable with acceptable response times, backed by performance screenshots.	Outstanding performance validation. Comprehensive stress testing data and clear metric screenshots prove the platform remains fast, stable, and resilient under a heavy simulated user surge.	
Scalability	Automatic scaling wasn't implemented at the application layer, leaving the system vulnerable to crashes under high load.	Automatic scaling was implemented at the application layer. However, scaling was significantly delayed or failed to trigger during peak traffic periods.	Automatic scaling was implemented at the application layer, but the configuration could be improved by adjusting the scaling policies, tracking metrics, or minimum/maximum capacities.	The automatic scaling configuration was optimal, staying strictly within the defined scaling range without hitting the maximum limit frequently. The application layer scaled in and out dynamically and in a timely manner according to live application demand.	

Assignment Rubrics (cont.)

Part 1: Reporting and Documentation (70 marks)

Criteria	Unsatisfactory (1-3)	Needs Improvement (4-5)	Good (6-8)	Excellent (9-10)	Mark
Load Balanced	Load balancing wasn't implemented. Traffic hits a single instance directly, creating a single point of failure.	Load balancing was implemented partially. Application traffic was directed to a single Availability Zone instead of utilizing multiple Availability Zones.	Application traffic was distributed across multiple Availability Zones, but configuration gaps caused instances to carry a variable or uneven load.	The application traffic was distributed evenly across multiple Availability Zones and across the instances within those zones. All backend instances were evenly loaded, managing peak user traffic seamlessly without performance degradation.	
Subtotal (Part 1)					

Assignment Rubrics (cont.)

Part 2: Demonstration Session and Student Effort (30 marks)

Criteria		Student Name (Block Capital)		
Presentation and Demonstration (15)	Clarity and Understanding (10)			
	Q&A (5)			
Student Effort (15)	Suggestion based on • Tutorial submission • Practical submission • Class Participation • Advanced Features			
Subtotal (Part 2)				
TOTAL (PART 1 + PART 2)				



Introduction

Project: SplashPoint — a cloud-based swimming pool booking system for TAR UMT Campus Aquatic Center

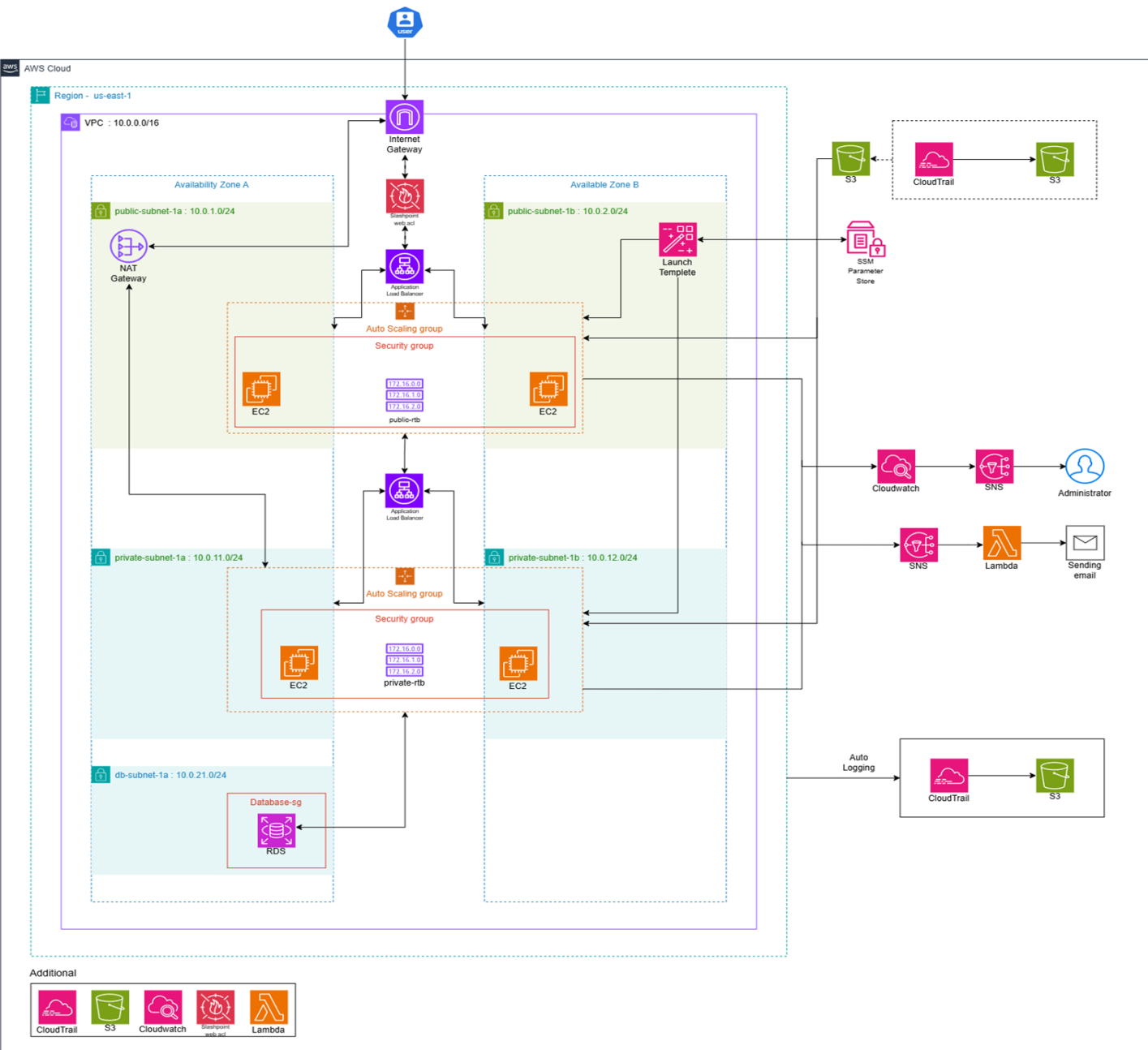
Problem: manual/legacy booking methods cause long queues, scheduling conflicts, and system overload during peak periods

Objective: deploy a scalable, secure, and cost-optimized booking platform as a proof of concept on AWS

Scope: 3 campus pools (1 Olympic-size lane pool, 2 leisure/training pools), public browsing, member accounts, real-time availability, admin management



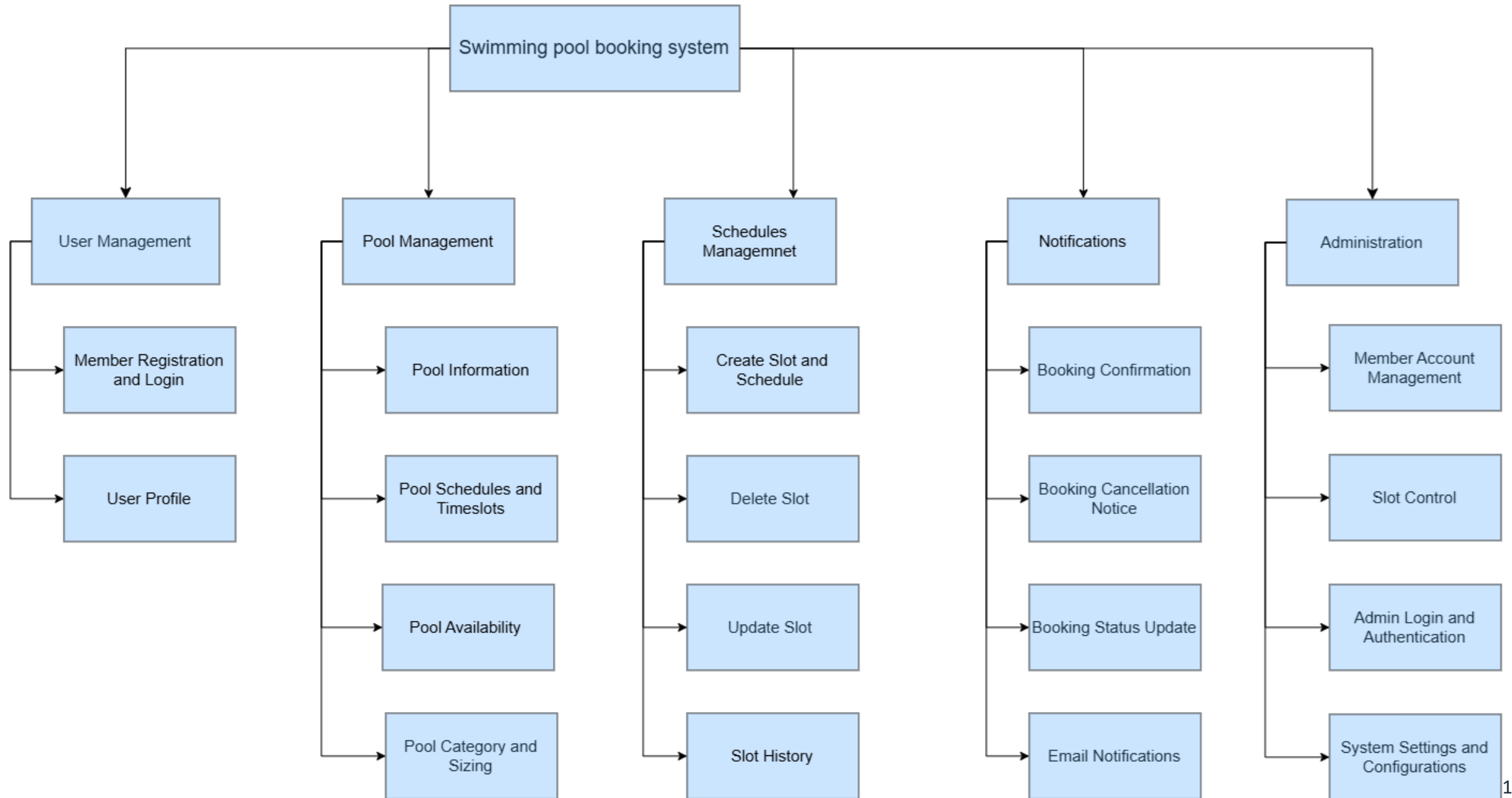
AWS Architecture Design





Solution Overview Diagram

Solution Overview Diagram





Key Features



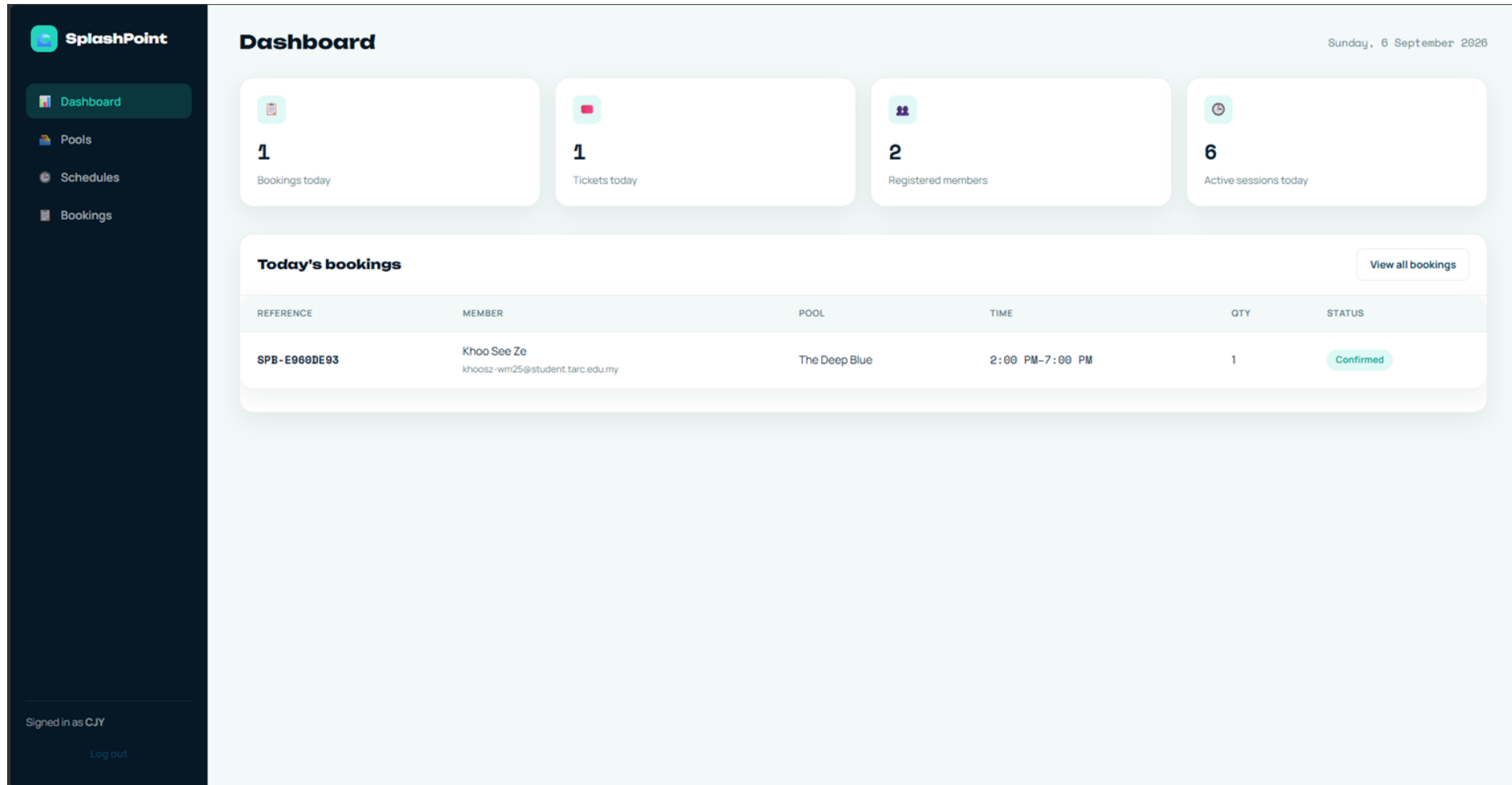
Functional

Key Features

Web site URL link

<http://splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com>

Admin Dashboard



Pools Management

SplashPoint

Dashboard

Pools

Schedules

Bookings

Manage Pools

Sunday, 6 September 2026

Add a new pool

Name

Slug (used for /images/ <slug>.jpg)

e.g. deep-blue

Tagline

Photo filename (in /images, or S3 if configured)

e.g. deep-blue.jpg

Description

Depth label

e.g. 1.2m - 2.0m

Capacity per session

20

Card size

Small

Accent color

Deep blue

Create pool

All pools

NAME	SIZE	CAPACITY	STATUS	
The Deep Blue deep-blue	Big	40	Active	EditDeactivate
Coral Cove coral-cove	Small	20	Active	EditDeactivate

19

Schedule Management



Dashboard

Pools

Schedules

Bookings

Signed in as CJY

Log out

Manage Schedules

Sunday, 6 September 2026

Add a recurring time slot

POOL

The Deep Blue

START TIME

--:--

END TIME

--:--

CAPACITY OVERRIDE

Uses pool default

Add slot

This defines a daily recurring time slot (e.g. every day 7:00-12:00) — it's not a one-off session. Actual bookable sessions for each date are generated automatically from this.

All recurring slots

POOL	TIME	CAPACITY	STATUS	
The Deep Blue	7:00 AM-12:00 PM	40 (default)	Active	PauseDelete
The Deep Blue	2:00 PM-7:00 PM	40 (default)	Active	PauseDelete
Coral Cove	8:00 AM-12:00 PM	20 (default)	Active	PauseDelete
Coral Cove	2:00 PM-6:00 PM	20 (default)	Active	PauseDelete
Tide Pool	8:00 AM-12:00 PM	20 (default)	Active	PauseDelete
Tide Pool	2:00 PM-6:00 PM	20 (default)	Active	PauseDelete

20

Booking Management

 **SplashPoint**

Dashboard

Pools

Schedules

Bookings

All Bookings

Sunday, 6 September 2026

Bookings

09/06/2026  **All** Confirmed Cancelled All dates

REFERENCE	MEMBER	POOL	SESSION	QTY	STATUS	
SPB-E968DE93	Khoo See Ze khoosz-wm25@student.tarc.edu.my · +601234567890	The Deep Blue	Today 2:00 PM-7:00 PM	1	Confirmed	Cancel

Signed in as CJY

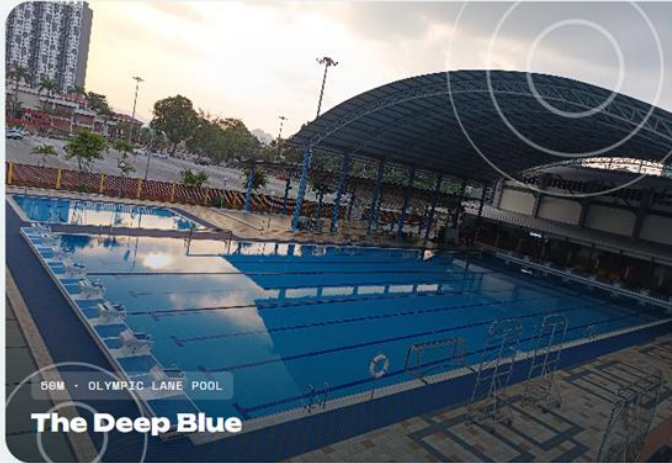
Log out

Check Availability



[Pools](#) [My Profile](#) [Log out](#)

Browse today's and tomorrow's sessions below. You can look around freely — you'll only need to log in when you're ready to book.



56M • OLYMPIC LANE POOL

The Deep Blue

Our full-length competition pool with 8 marked lanes. Ideal for serious lap swimmers, squad training, and anyone who wants real distance.

DEPTH	CAPACITY	NEXT SESSION
1.2m - 2.0m	40 people	Today 2:00 PM

0%

NEXT SESSION

40 spots left

FREE

Today • 2:00 PM-7:00 PM

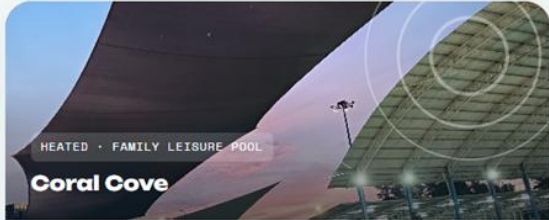
40 left

Tomorrow • 7:00 AM-12:00 PM

40 left

Tomorrow • 2:00 PM-7:00 PM

40 left



HEATED • FAMILY LEISURE POOL

Coral Cove

A warm, shallow pool built for families and casual swimmers. Gentle steps, no strong currents, and plenty of room to relax.



QUIET • TRAINING POOL

Tide Pool

A compact 4-lane pool for focused technique work, rehab swimming, or anyone who prefers a quieter session away from the crowds.

Booking



[Pools](#) [My Profile](#) [Log out](#)

Book: The Deep Blue

Today · 7:00 AM-12:00 PM | 40 spots left | Free

Booking as JUN YU CHENG (chengjy-wm25@student.tarc.edu.my)

Tickets 1 (one booking per person)

Cost Free 🎉

Confirm Booking

Cancel

Book Successful



Booking Confirmed!

Thank you, JUN YU CHENG. Your reservation is set.

SPB-1C28F32C

The Deep Blue

Today

7:00 AM – 12:00 PM

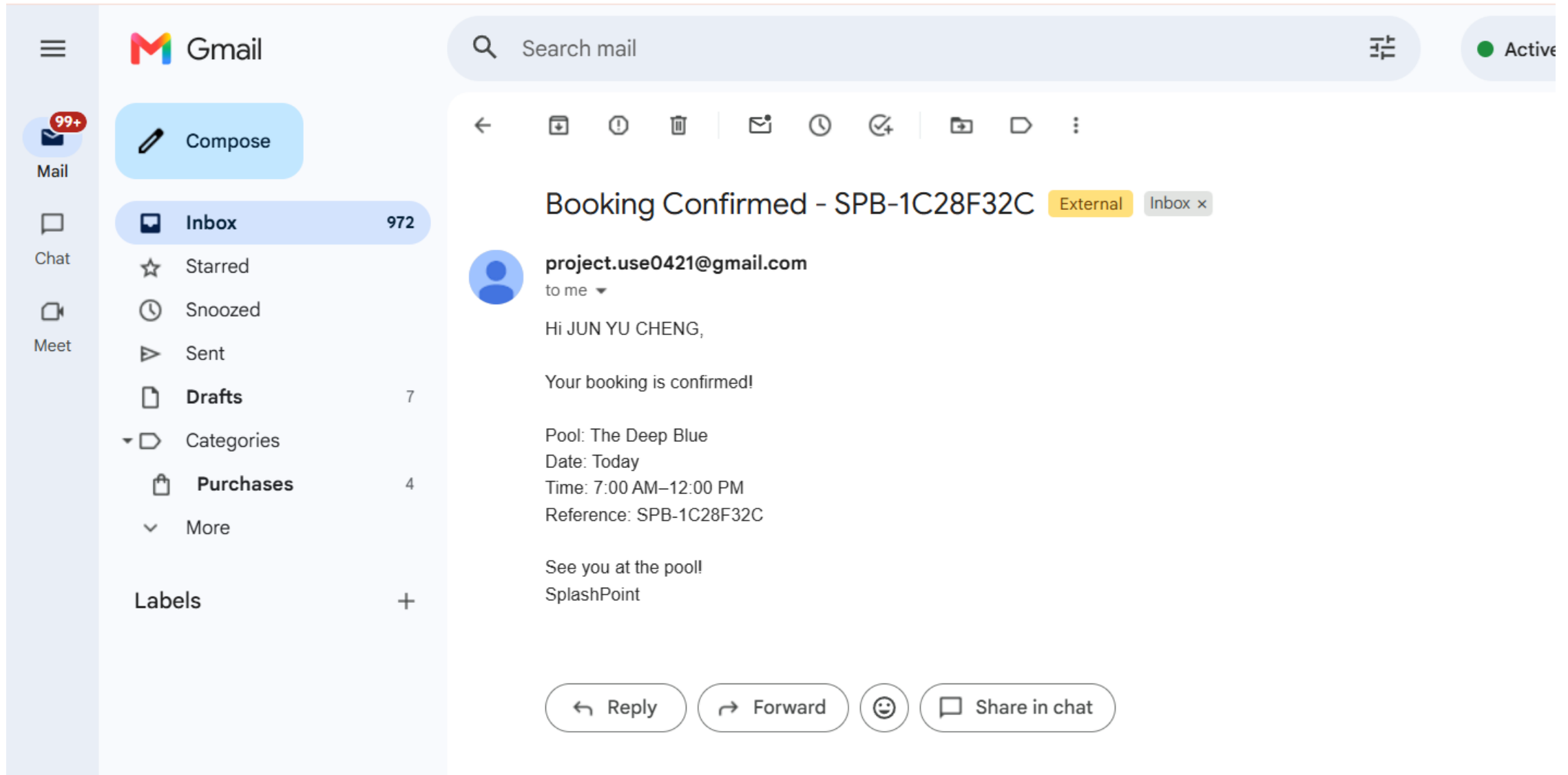
Tickets: 1 | Cost: Free

A confirmation has been recorded under chengjy-wm25@student.tarc.edu.my. You can find this booking any time in your profile.

[Book Another Session](#)

[View My Bookings](#)

After book successful, will receive a email



Profile & View / Cancel Booking

Hi, JUN YU CHENG

Your details

Full name

JUN YU CHENG

Username (used to log in)

CJY

Email address

chengjy-wm25@student.tarc.edu.my

Phone number

011-28003667

Save changes

Your bookings

REFERENCE	POOL	SESSION	QTY	STATUS
SPB-1C28F32C	The Deep Blue	Today · 7:00 AM-12:00 PM	1	Confirmed Cancel



Secure

Key Features



Secure

Secure — Network & Data Protection



Secure

Key Features - System Manager Services

vpc-0fe59c5f43ae9ef53 / splashpoint-vpc

Actions

Details Info

vpc-0fe59c5f43ae9ef53

Enabled

acl-03fc0c45504e0ffd9

100

1000

Available

default

No

Disabled

—

⊖ Off

dopt-02533e501b5c33652

10.0.0.0/16

groups
⊗ Failed to load rule groups

Enabled

rtb-051e60f54b8bde450

11

082810551484

Resource map

CIDRs

Flow logs

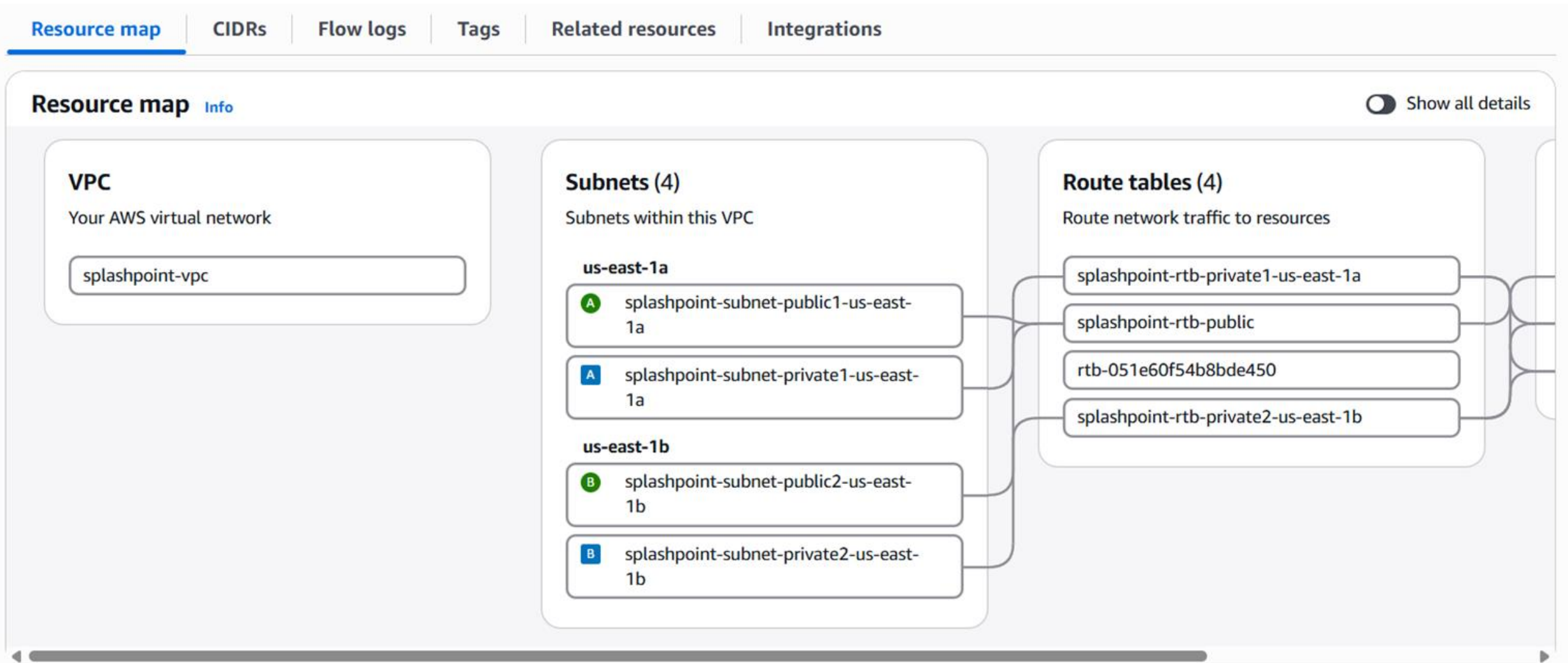
Tags

Related resources

Integrations

Resource map [Info](#)☐ Show all details

2 AZ - 2 Public & 2 Private Subnet



NAT Gateway & Elastic IP

nat-0bd8f749c3f3dbf66



nat-0bd8f749c3f3dbf66 / splashpoint-nat-gateway

Actions ▼

Details

NAT gateway ID

nat-0bd8f749c3f3dbf66

NAT gateway ARN

arn:aws:ec2:us-east-1:082810551484:natgateway/nat-0bd8f749c3f3dbf66

VPC

vpc-0fe59c5f43ae9ef53 / splashpoint-vpc

Connectivity type

Public

Primary public IPv4 address

44.213.11.48

Subnet

subnet-026297abbd88bd23a / splashpoint-subnet-public1-us-east-1a

State

Available

Primary private IPv4 address

10.0.0.253

Created

Sunday, September 6, 2026 at 17:28:46 GMT+8

State message [Info](#)

–

Primary network interface ID

eni-05f3d70e5d566c99a

Deleted

–

Elastic IP addresses (3) [Info](#)



Actions ▼

Allocate Elastic IP address

Find elastic IP addresses by attribute or tag

< 1 >

☐	Name ▼	Allocated IPv4 addr... ▼	Type ▼	Allocation ID ▼	Reverse DNS record ▼	Assoc
☐	–	100.60.114.42	Public IP	eipalloc-079306fd3ddf96cc4	–	–
☐	–	18.215.206.205	Public IP	eipalloc-048dc429c96b22da2	–	–
☐	–	44.213.11.48	Public IP	eipalloc-001dbfa1cb223407f	–	–



Route Table

rtb-05db24f76164d431f / splashpoint-rtb-private1-us-east-1a

Actions ▾

Details [Info](#)

Route table ID

 rtb-05db24f76164d431f

VPC

 vpc-0fe59c5f43ae9ef53 | splashpoint-vpc

Main

 No

Owner ID

 082810551484

Explicit subnet associations

[subnet-0f7fa97b0e9549a81 / splashpoint-subnet-private1-us-east-1a](#)

Edge associations

–

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (3)

Both ▾

Edit routes

 Filter routes

< 1 > 

Destination ▾	Target ▾	Status ▾	Propagated ▾	Route Origin ▾
pl-63a5400a	vpce-09638ecd9dbec28af	Active	No	Create Route
0.0.0.0/0	nat-0bd8f749c3f3dbf66	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

rtb-084097fd270be5259 / splashpoint-rtb-private2-us-east-1b

Actions ▾

Details [Info](#)

Route table ID

 rtb-084097fd270be5259

VPC

 vpc-0fe59c5f43ae9ef53 | splashpoint-vpc

Main

 No

Owner ID

 082810551484

Explicit subnet associations

[subnet-03523195238654009 / splashpoint-subnet-private2-us-east-1b](#)

Edge associations

–

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (3)

Both ▾

Edit routes

 Filter routes

< 1 > 

Destination ▾	Target ▾	Status ▾	Propagated ▾	Route Origin ▾
pl-63a5400a	vpce-09638ecd9dbec28af	Active	No	Create Route
0.0.0.0/0	nat-0bd8f749c3f3dbf66	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Security Group - sg-alb

EC2 > Security Groups > Create security group

Basic details

Security group name

Info

splashpoint-sg-alb

Name cannot be edited after creation.

Description

Info

splashpoint-sg-alb

VPC

Info

vpc-0fe59c5f43ae9ef53 (splashpoint-vpc)

Inbound rules

Info

Type	Info	Protocol	Info	Port range	Info	Source	Info		Description - optional	Info	
HTTP		TCP		80		Anywh...		0.0.0.0/0 X			Delete
HTTPS		TCP		443		Anywh...		0.0.0.0/0 X			Delete

Add rule

Security Group - sg-alb

sg-0f099ba16571d906f

sg-0f099ba16571d906f - splashpoint-sg-alb

Actions

Details

Security group name

splashpoint-sg-alb

Security group ID

sg-0f099ba16571d906f

Description

splashpoint-sg-alb

VPC ID

[vpc-0fe59c5f43ae9ef53](#)

Owner

082810551484

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-09c65ef183713a938	IPv4	HTTPS	TCP	443	0.0.0.0/0
sgr-090d50378180d8951	IPv4	HTTP	TCP	80	0.0.0.0/0

Security Group - sg-ec2

EC2 > Security Groups > Create security group

Basic details

Security group name

Info

splashpoint-sg-ec2

Name cannot be edited after creation.

Description

Info

splashpoint-sg-ec2

VPC

Info

vpc-0fe59c5f43ae9ef53 (splashpoint-vpc)

Inbound rules

Info

Type	Info	Protocol	Info	Port range	Info	Source	Info	
HTTP		TCP		80		Custom		Use: "alb" Security Groups splashpoint-sg-alb sg-0f099ba16571d906f alb CIDR block, a security group ID or a prefix list has to be specified. 42.188.132.106/32 Delete
SSH		TCP		22		My IP		Delete

Add rule

36

Security Group - sg-ec2

sg-02ed0eb567b6aa3aa

sg-02ed0eb567b6aa3aa - splashpoint-sg-ec2

Actions ▼

Details

Security group name ☐ splashpoint-sg-ec2	Security group ID ☐ sg-02ed0eb567b6aa3aa	Description ☐ splashpoint-sg-ec2	VPC ID vpc-0fe59c5f43ae9ef53 ↗
Owner ☐ 082810551484	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

- Inbound rules
- Outbound rules
- Sharing
- VPC associations
- Related resources - new
- Tags

Inbound rules (2)

Manage tags Edit inbound rules

Security group rule ID ▼	IP version ▼	Type ▼	Protocol ▼	Port range ▼	Source ▼	Des
sgr-06e5a50906418519f	-	HTTP	TCP	80	sg-0f099ba16571d906f...	-
sgr-0a6e5c92543fd85	IPv4	SSH	TCP	22	210.195.116.211/32	-

Security Group - sg-rds

EC2 > Security Groups > Create security group

EC2 > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, you must specify a name, description, and VPC.

Basic details

Security group name

Info

splashpoint-sg-rds

Name cannot be edited after creation.

Description

Info

splashpoint-sg-rds

VPC

Info

vpc-0fe59c5f43ae9ef53 (splashpoint-vpc)

Inbound rules

Info

Type

Info

Protocol

Info

Port range

Info

Source

Info

Description - optional

Info

Delete

MYSQL/Aurora

TCP

3306

Custom

Use: "ec2"

CIDR blocks

ec2::/16

ec2::/32

ec2::/48

ec2::/64

ec2::/80

ec2::/96

ec2::/112

ec2::/128

Security Groups

splashpoint-sg-ec2 | sg-02ed0eb567b6aa3aa

ec2

CIDR block, a security group ID or a prefix list has to be specified.

38

Security Group - sg-rds

sg-066d78735ad1451a6



sg-066d78735ad1451a6 - splashpoint-sg-rds

Actions ▾

Details

Security group name ☐ splashpoint-sg-rds	Security group ID ☐ sg-066d78735ad1451a6	Description ☐ splashpoint-sg-rds	VPC ID vpc-0fe59c5f43ae9ef53
Owner ☐ 082810551484	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

- Inbound rules
- Outbound rules
- Sharing
- VPC associations
- Related resources - new
- Tags

Inbound rules (1)

Manage tags Edit inbound rules

Search

< 1 >

▼	Security group rule ID ▼	IP version ▼	Type ▼	Protocol ▼	Port range ▼	Source ▼
	sgr-041cc02492b322916	-	MYSQL/Aurora	TCP	3306	sg-02ed0eb567b6aa3a...

RDS

splashpoint-db



splashpoint-db



Modify

Actions ▼

Summary

DB identifier splashpoint-db	Status ✔ Available	Role Instance	Engine MySQL Community	Recommendations
CPU 5.99%	Class db.t3.micro	Current activity 0 Connections	Region & AZ us-east-1a	

- <
- Connectivity & security
- Monitoring
- Logs & events
- Configuration
- Zero-ETL integrations
- Maintenance & backups
- Data
- >

Connect using [Info](#)

- ☐ Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.
- ☐ CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
- ☒ Endpoints
Use when connecting through any IDE interface.

Database name  pool_booking	Master username  admin	Internet access gateway  Disabled	Port  3306
--	---	--	---

Endpoint type

Instance endpoint ▼

RDS

Connectivity & security

Endpoint & port

Endpoint
 splashpoint-db.cbzsvuio0uqx.us-east-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-1a

VPC
[splashpoint-vpc \(vpc-0fe59c5f43ae9ef53\)](#)

Subnet group
default-vpc-0fe59c5f43ae9ef53

Subnets
[subnet-03523195238654009](#)
[subnet-0f7fa97b0e9549a81](#)
[subnet-026297abbd88bd23a](#)
[subnet-0f27dccf81d9722c8](#)

Network type
IPv4

Security

VPC security groups
[splashpoint-sg-rds \(sg-066d78735ad1451a6\)](#)
 Active

Publicly accessible
No

Certificate authority [Info](#)
rds-ca-rsa2048-g1

Certificate authority date
May 26, 2061, 07:34 (UTC+08:00)

DB instance certificate expiration date
September 06, 2027, 20:29 (UTC+08:00)

Private EC2

i7e4582f08a0906

Instance summary for i-07b7e4582f08a0906 (splashpoint-ec2) [Info](#)

Updated less than a minute ago

Instance ID

 i-07b7e4582f08a0906

IPv6 address

–

Hostname type

IP name: ip-10-0-135-96.ec2.internal

Answer private resource DNS name

–

Auto-assigned IP address

–

IAM role

 LabRole [↗](#)

IMDSv2

Required

Public IPv4 address

–

Private IPv4 addresses

 10.0.135.96

Instance state

 Running

Private IP DNS name (IPv4 only)

 ip-10-0-135-96.ec2.internal

Instance type

t3.micro

VPC ID

 vpc-0fe59c5f43ae9ef53 (splashpoint-vpc) [↗](#)

Subnet ID

 subnet-0f7fa97b0e9549a81 (splashpoint-subnet-private1-us-east-1a) [↗](#)

Instance ARN

 arn:aws:ec2:us-east-1:082810551484:instance/i-07b7e4582f08a0906

Public DNS

–

Elastic IP addresses

–

AWS Compute Optimizer finding

 Opt-in to AWS Compute Optimizer for recommendation S. [Learn more](#) [↗](#)

Auto Scaling Group name

–

Managed

false

Our security design follows the "zero trust" principle—the EC2 application layer and the RDS database layer are completely isolated in a private subnet, with zero exposure to the external network. The only external entry point is ALB, with layers of restrictions from Security Group: ALB only accepts HTTP/HTTPS, EC2 only trusts traffic from ALB, and RDS only trusts traffic from EC2, forming a one-way trust chain.



Secure

Secure — Secrets Management & Edge
Protection



Secure

Key Features - System Manager



/pool-booking/db-host

Edit Delete

- Overview
- History
- Tags

Parameter details

Name /pool-booking/db-host	Description Save DB endpoint host
ARN  arn:aws:ssm:us-east-1:082810551484:parameter/pool-booking/db-host	Data type text
Tier Standard	Last modified user arn:aws:sts::082810551484:assumed-role/voclabs/user5166388=chengjy-wm25@student.tarc.edu.my
Type String	Last modified date Sun, 06 Sep 2026 12:32:22 GMT
Value splashpoint-db.cbzsvuio0uqx.us-east-1.rds.amazonaws.com	Version 6



/pool-booking/db-pass

Edit
 Delete

Overview
 History
 Tags

Parameter details

Name
 /pool-booking/db-pass

ARN
 arn:aws:ssm:us-east-1:082810551484:parameter/pool-booking/db-pass

Tier
 Standard

Type
 SecureString

Value
 Show decrypted value

Description
 Save db password

Data type
 text

Last modified user
 arn:aws:sts::082810551484:assumed-role/voclabs/user5166388=chengjy-wm25@student.tarc.edu.my

Last modified date
 Tue, 25 Aug 2026 07:10:04 GMT

Version
 1

Value
 Show decrypted value
 12345678

Key Features - Secure

☰

[AWS Systems Manager](#) > [Parameter Store](#)

ⓘ

🖨

Parameter Store

My parameters

Public parameters

Settings

ⓘ Ready to scale? Enable higher throughput to handle more API calls - Manage Settings to increase TPS.

Manage settings

My parameters

🔍 Filter parameters

🔄

View details

Edit

Delete

Create parameter

< 1 > ⚙

☐	Name ▾	Tier ▾	Type ▾	Last modified ▾
☐	/pool-booking/db-host	Standard	String	Sun, 06 Sep 2026 12:32:22 GMT
☐	/pool-booking/db-name	Standard	String	Wed, 02 Sep 2026 08:29:51 GMT
☐	/pool-booking/db-pass	Standard	SecureString	Tue, 25 Aug 2026 07:10:04 GMT
☐	/pool-booking/db-user	Standard	String	Wed, 02 Sep 2026 08:29:06 GMT

```
DB_HOST=$(aws ssm get-parameter --name "/pool-booking/db-host" --region us-east-1 --query "Parameter.Value" --output text)
DB_USER=$(aws ssm get-parameter --name "/pool-booking/db-user" --region us-east-1 --query "Parameter.Value" --output text)
DB_PASS=$(aws ssm get-parameter --name "/pool-booking/db-pass" --with-decryption --region us-east-1 --query "Parameter.Value" --output text)
DB_NAME=$(aws ssm get-parameter --name "/pool-booking/db-name" --region us-east-1 --query "Parameter.Value" --output text)
```

```
sudo tee /etc/httpd/conf.d/db-env.conf << EOF
SetEnv DB_HOST $DB_HOST
SetEnv DB_USER $DB_USER
SetEnv DB_PASS $DB_PASS
SetEnv DB_NAME $DB_NAME
SetEnv IMAGE_BASE_URL https://jy-pool-booking-bucket.s3.us-east-1.amazonaws.com/
EOF
```



Secure

Key Features - WAF & Shield

WAF - Web Application Firewall

WAF & Shield > Protection packs (web ACLs) > splashpoint-web-acl

Resources and protection packs (web ACLs)

View your secured assets and their active security controls.

Protection packs (web ACLs) (1) Info

Use protection packs (web ACLs) to combine rules and rule sets that protect your infrastructure from threats.

Search protection packs (web ACLs)

Region scope

CloudFront (Global) and Regional

< 1 >

List

Grid

Name

Rules

Resources

Dashboard

AI traffic analysis [new](#)

Sampled request logs

Log events

splashpoint-web-acl

2 rules

Manage

View

View

View

Manage resources

< splashpoint-web-acl

☐ splashpoint-alb Application Load Balancer

Resource level DDoS protection

Protect critical resources from DDoS attacks using AWS managed threat list. This configuration currently only applies to Application Load Balancers.

Protection against low reputation sources

Active under DDoS 

Cancel

Disassociate

Add regional resources

WAF & Shield > Protection packs (web ACLs) > splashpoint-web-acl

Resources and protection packs (web ACLs)

View your secured assets and their active security controls.

Protection packs (web ACLs) (1) Info

Use protection packs (web ACLs) to combine rules and rule sets that protect your infrastructure from threats.

Search protection packs (web ACLs)

Region scope

CloudFront (Global) and Regional

< 1 >

List

Grid

Name	Rules	Resources	Dashboard	AI traffic analysis new	Sampled request logs	Log events
splashpoint-web-acl	3 rules	Manage	View	View	View	

Manage rules for splashpoint-web-acl

< splashpoint-web-acl

Rules are shown in ascending rule priority order.

AWS-AWSManagedRulesCommonRuleSet 700 WCU >

AWS-AWSManagedRulesSQLiRuleSet 200 WCU >

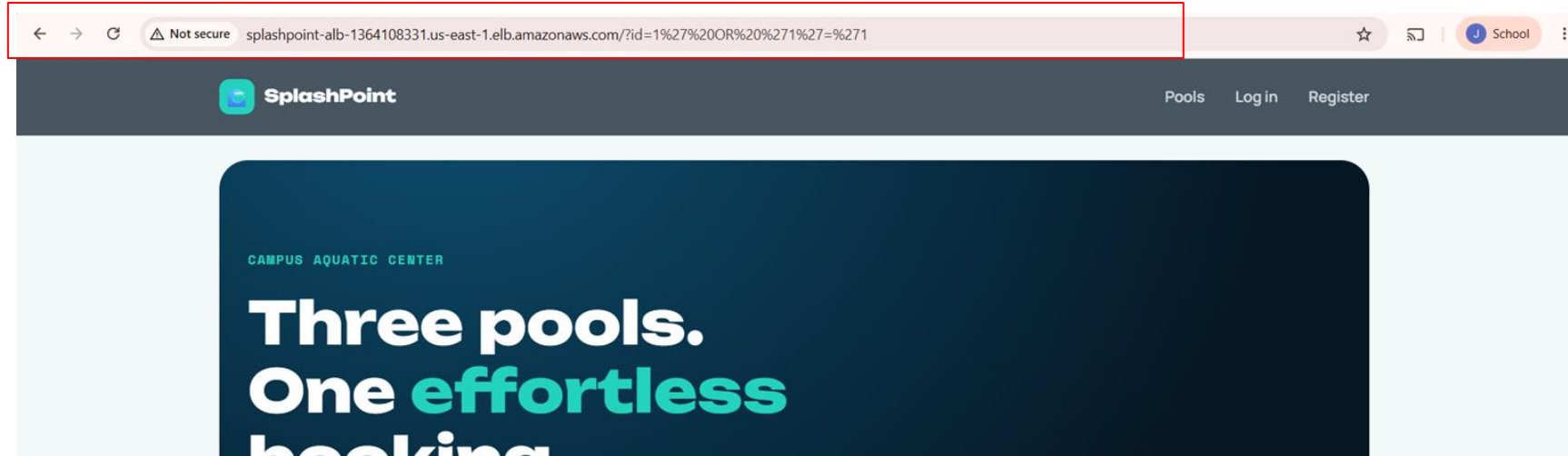
AWS-AWSManagedRulesAmazonIpReputationList 25 WCU >

Add rule

Cancel

Edit rule order

When did not have setup WAF



Simulated attack test add - ?id=1' OR '1'='1



In addition to network layer isolation, we also implemented application layer and edge layer protection. Database passwords are not hard-coded in configuration files, but are encrypted and stored using SecureString in the SSM Parameter Store, and EC2 dynamically retrieves them at boot time via IAM Role. Member passwords use bcrypt one-way hashes; the database does not store plaintext, so even if data is leaked, passwords cannot be reverse-engineered. An outermost layer of WAF is added to intercept common attacks such as SQL injection and XSS. We have actually tested attacks with malicious payloads; the system correctly returned a 403 error and recorded it.



Cost Optimized

Key Features

Summary report link:
<https://calculator.aws/#/estimate?id=23588b3280df473ebb580ed19157a1e86458921e>

Total Cost need : 111.59 USD per month



Contact your AWS representative: [Contact Sales](#)

Export Date: 09/07/2026

Language: English

[Estimate url](#)

Estimate summary		
Upfront cost 0.00 USD	Monthly cost 111.59 USD	Total 12 months cost 1,339.08 USD Includes upfront cost

Detailed Estimate

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	16.78 USD
Status	-			
Description:	Amazon EC2			
Config summary	Tenancy (Shared Instances), Operating system (Linux), Workload (Consistent, Number of instances: 2), Advance EC2 instance (t3.micro), Pricing strategy (On-Demand Utilization: 100 %Utilized/Month), Enable monitoring (disabled), EBS Storage amount (8 GB), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)			

This EC2 configuration is designed for a small application that needs to run continuously while keeping the overall cost low. Shared tenancy is selected because the physical server resources are shared with other AWS customers, making it a more affordable option. Linux is chosen because it is lightweight, widely supported, and suitable for many types of applications. Two t3.micro instances are used to provide basic reliability, so the application can continue operating if one instance experiences a problem. On-Demand pricing is suitable because the instances are expected to run continuously without requiring a long-term commitment. The 8 GB EBS storage provides basic space for the operating system and application files, while monitoring is disabled to avoid additional costs. The data transfer values are set to zero because the estimate assumes very little network traffic. Overall, this configuration provides a simple, affordable, and reliable solution for a small workload.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Amazon RDS for MySQL	-	US East (N. Virginia)	0.00 USD	36.61 USD
Status	-			
Description:	-			
Config summary	Storage amount (20 GB), Storage for each RDS instance (General Purpose SSD (gp2)), Nodes (1), Instance type (db.t3.micro), Utilization (On-Demand only) (100 %Utilized/Month), Deployment option (Single-AZ), Pricing strategy (OnDemand)			

This RDS for MySQL configuration is designed for a small database workload that needs to run continuously while keeping costs low. The 20 GB storage provides enough space for a small application database, while General Purpose SSD (gp2) offers a balance between storage performance and cost. One RDS node is selected because the workload is small and does not require a standby database. The db.t3.micro instance is a low-cost option suitable for applications with light database usage. 100% utilization per month means the database is expected to run continuously, 24 hours a day. The Single-AZ deployment is chosen because it is cheaper and suitable when high availability is not a major requirement. Finally, On-Demand pricing provides flexibility because there is no long-term commitment. Overall, this configuration provides a simple and affordable MySQL database solution for a small application.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Elastic Load Balancing	-	US East (N. Virginia)	0.00 USD	16.63 USD
Status	-			
Description:	-			
Config summary	Number of Application Load Balancers (1)			

The Elastic Load Balancing configuration with one Application Load Balancer (ALB) is suitable for distributing incoming application traffic across the two EC2 instances. The load balancer helps ensure that requests are directed to available servers, improving the application's reliability and availability. Using one ALB is sufficient for this small application setup and helps keep the overall AWS cost lower.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Amazon Virtual Private Cloud (VPC)	-	US East (N. Virginia)	0.00 USD	33.30 USD
Status	-			
Description:	-			
Config summary	Working days per month (22) Number of Regional NAT Gateways (5), Number of Availability Zones Regional NAT Gateways is active in (3), Number of NAT Gateways (1)			

This Amazon VPC configuration is designed to provide secure internet connectivity for the application while controlling costs. The 22 working days per month represents the estimated period of usage during a typical month. One NAT Gateway is used to allow resources in private subnets to access the internet securely without being directly exposed to incoming internet traffic. The NAT Gateway is active across 3 Availability Zones, providing better network availability. The configuration also includes 5 regional NAT Gateways, which represents the estimated regional usage for the cost calculation. Overall, this setup provides a balance between security, availability, and cost for the application.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Amazon Simple Storage Service (S3)	-	US East (N. Virginia)	0.00 USD	0.17 USD
Status	-			
Description:	-			
Config summary	S3 Standard storage (5 GB per month), PUT, COPY, POST, LIST requests to S3 Standard (10000), GET, SELECT, and all other requests from S3 Standard (10000) DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month)			

This Amazon S3 configuration is designed for storing a small amount of application data at a low cost. The 5 GB of S3 Standard storage is suitable for storing files, images, documents, backups, or other application data that may need frequent access. The 10,000 PUT, COPY, POST, and LIST requests and 10,000 GET and other requests represent a small amount of activity between the application and S3. Data transfer is set to 0 TB because the estimate assumes there is no significant data transferred in or out of S3. Overall, this configuration provides simple, scalable, and affordable cloud storage for a small application.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
AWS Web Application Firewall (WAF)	-	US East (N. Virginia)	0.00 USD	7.00 USD
Status	-			
Description:	-			
Config summary	Number of Web Access Control Lists (Web ACLs) utilized (1 per month), Number of Rules added per Web ACL (2 per month)			

The AWS Web Application Firewall (WAF) configuration is designed to protect the web application from common online threats. One Web ACL is used to control and filter incoming web traffic, while two rules are added to determine which requests should be allowed or blocked. This provides basic security for the application while keeping the configuration simple and cost-effective. Overall, AWS WAF helps protect the application from malicious or unwanted web traffic.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
AWS Lambda	-	US East (N. Virginia)	0.00 USD	0.00 USD
Status	-			
Description:	-			

Config summary

Architecture (x86), Architecture (x86), Invoke Mode (Buffered), Amount of ephemeral storage allocated (512 MB), Number of requests (10000 per month)

This AWS Lambda configuration is designed for a small application that needs to run code without managing servers directly. The x86 architecture is chosen because it provides broad compatibility with common applications and libraries. The Buffered invoke mode allows Lambda to process requests asynchronously, which is suitable for tasks that do not require an immediate response. The 512 MB ephemeral storage provides temporary space for processing files and data during execution. With 10,000 requests per month, this configuration is suitable for a light workload while keeping costs low. Overall, AWS Lambda provides a flexible and cost-effective way to run application functions without maintaining additional servers.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Amazon Simple Notification Service (SNS)	-	US East (N. Virginia)	0.00 USD	0.00 USD
Status	-			
Description:	-			
Config summary	DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), HTTP/HTTPS Notifications (1000 per month), EMAIL/EMAIL-JSON Notifications (1000 per month)			

The Amazon Simple Notification Service (SNS) configuration is used to send notifications and messages between different parts of the application. The HTTP/HTTPS notifications (1,000 per month) allow SNS to send messages to web applications or services, while EMAIL/EMAIL-JSON notifications (1,000 per month) allow notifications to be delivered through email. Data transfer is set to 0 TB because the estimate assumes there is no significant additional data transfer. Overall, SNS provides a simple and reliable way to send alerts, updates, and application notifications while keeping the setup cost low.

Key Features - Cost Optimized

Name	Group	Region	Upfront cost	Monthly cost
Amazon CloudWatch	-	US East (N. Virginia)	0.00 USD	1.10 USD
Status	-			
Description:	-			
Config summary	Number of mobile OTEL events and spans or spans per visit (70), Mobile sampling rate (1), Number of Metrics (includes detailed and custom metrics) (3), Number of Dashboards (1), Number of Standard Resolution Alarm Metrics (2)			

The Amazon CloudWatch configuration is used to monitor the performance and health of the application. The 70 mobile OTEL events and spans are used to collect application activity and performance data, while a sampling rate of 1 means the monitoring collects all the specified events. Three metrics are used to track important application information, and one dashboard provides a central place to view the system's performance. Two standard-resolution alarm metrics are used to detect potential problems and generate alerts when certain conditions occur. Overall, this configuration provides basic monitoring and alerting to help maintain the application's performance and reliability.

The following components and management services implemented in the architecture incur \$0.00 base charge or fall entirely within the AWS Free Tier, and are categorized as **Free / Negligible Cost** in the estimate:

- **AWS Systems Manager (Parameter Store):** Standard Parameters usage is free.
- **AWS CloudTrail:** The first copy of Management Events (1st Trail) per region is free.
- **AWS Auto Scaling Group:** The Auto Scaling service itself has no additional cost; charges apply only to the underlying EC2 instances it provisions.
- **Amazon VPC & Security Groups:** Virtual Private Cloud infrastructure, subnets, routing tables, and Security Group network rules are provided free of charge.



High Performing

Key Features



High Performing

Key Features - CloudWatch Alarm

Test EC2-High-CPU-Alarm

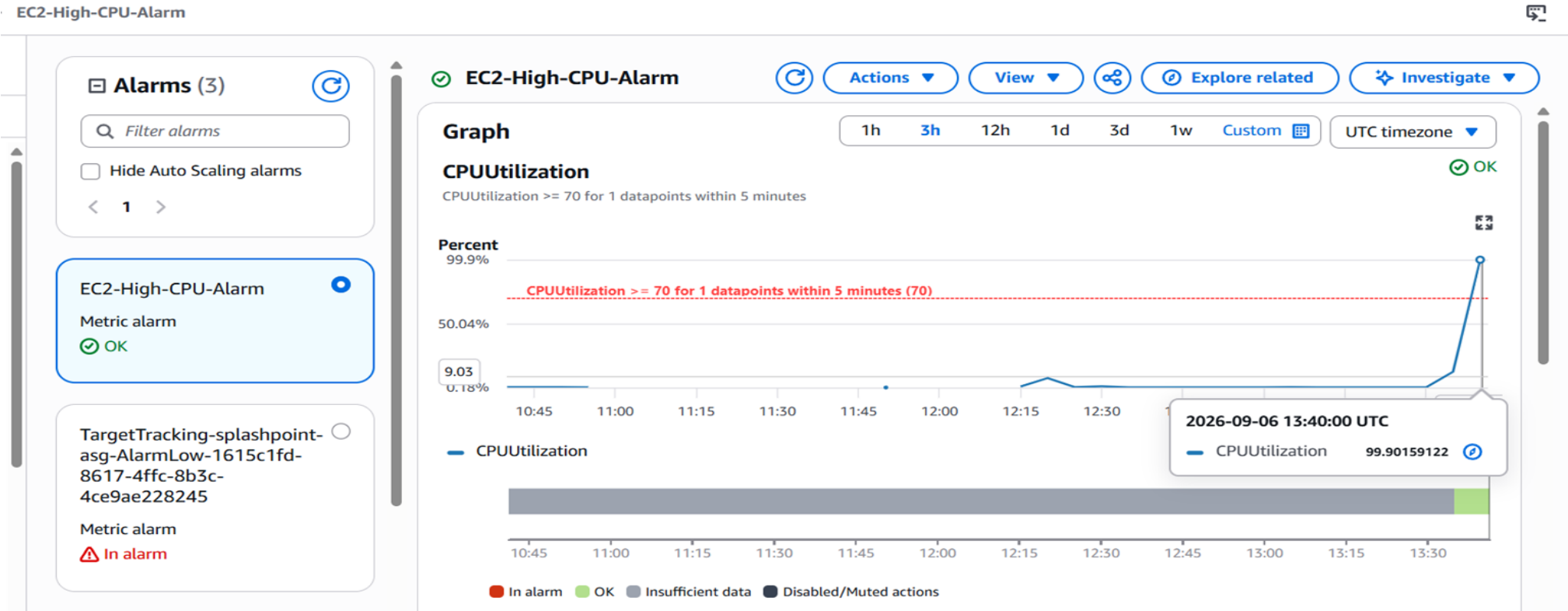
Session ID: user5166388=chengjy-wm25@student.tarc.edu.my-y3jecfpa23g9krvso263h2ckh8

Shortcuts

Instance ID: i-07b7e4582f08a0906 (splashpoint-ec2)

Terminate session

```
sh-5.2$ yes > /dev/null &
[1] 5453
sh-5.2$ yes > /dev/null &
[2] 5457
sh-5.2$
```



Email notification

ALARM: "EC2-High-CPU-Alarm" in US East (N. Virginia)

ExternalInbox x

CloudWatch_Alarms_Topic <no-reply@sns.amazonaws.com>

9:41PM (0 minutes ago)



to me ▼

You are receiving this email because your Amazon CloudWatch Alarm "EC2-High-CPU-Alarm" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [99.90159121993528 (06/09/26 13:36:00)] was greater than or equal to the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Sunday 06 September, 2026 13:41:57 UTC".

View this alarm in the AWS Management Console:

<https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/EC2-High-CPU-Alarm>

Alarm Details:

- Name: EC2-High-CPU-Alarm
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [99.90159121993528 (06/09/26 13:36:00)] was greater than or equal to the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Sunday 06 September, 2026 13:41:57 UTC
- AWS Account: 082810551484
- Alarm Arn: arn:aws:cloudwatch:us-east-1:082810551484:alarm:EC2-High-CPU-Alarm

Threshold:

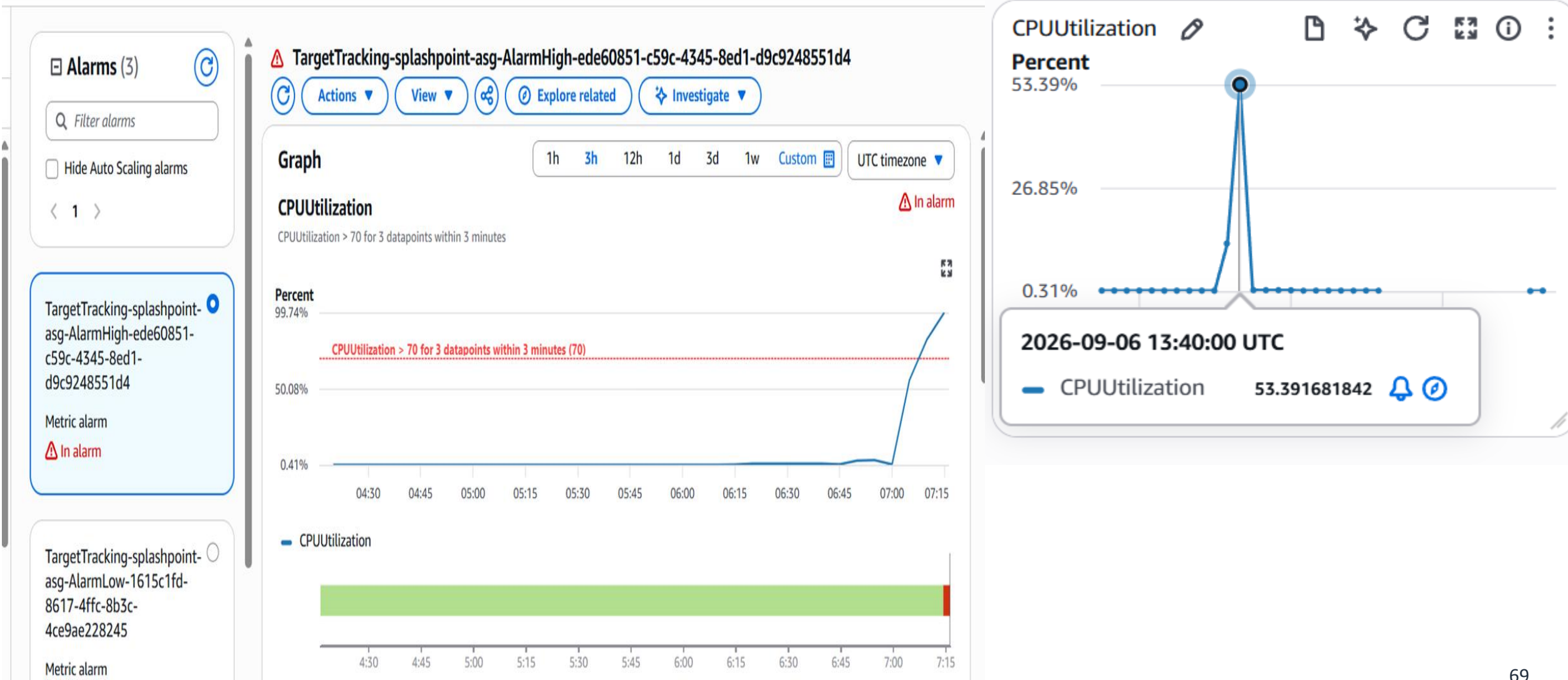
- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 70.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: AWS/EC2

Asg - alarm

TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4

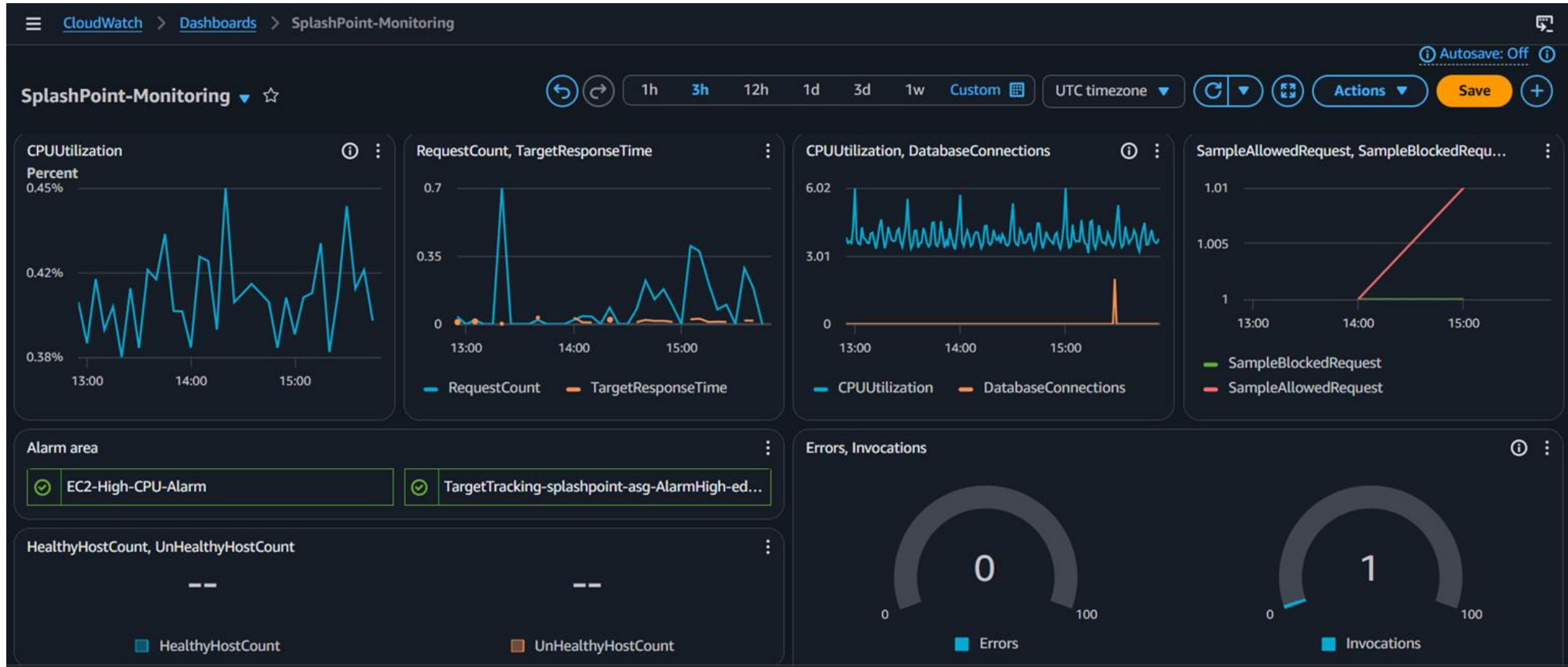




High Performing

Key Features - CloudWatch Dashboard

CloudWatch Dashboard + Alarm





High Performing

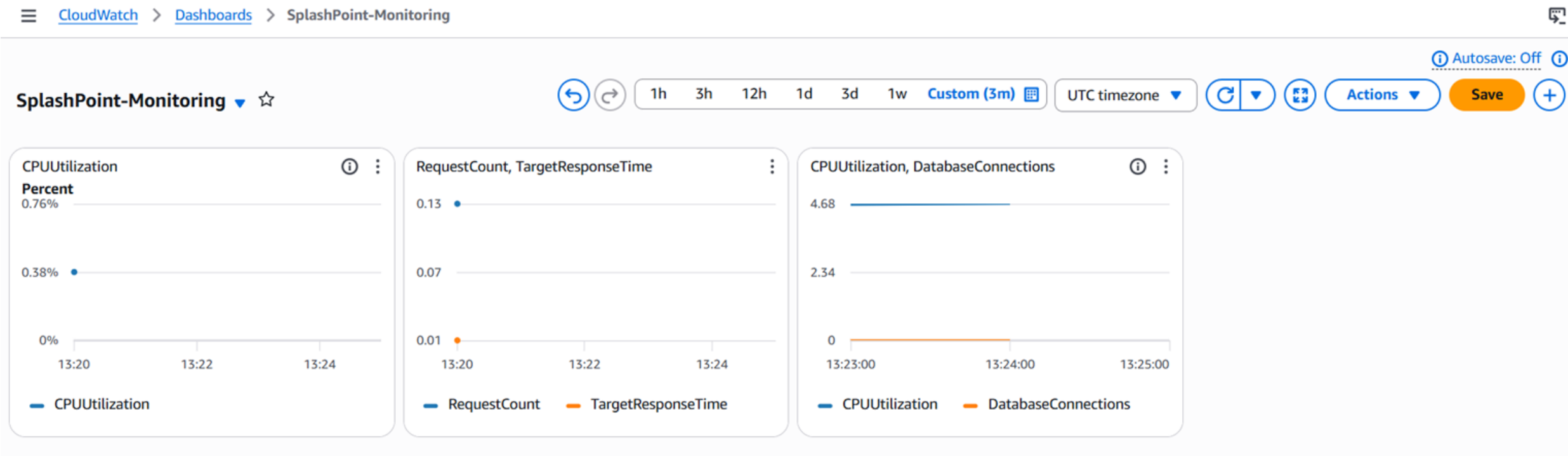
Key Features - Test Setup & Execution

We used Apache Bench to simulate 50 concurrent users making a total of 50000 requests to the homepage. During the test, there were zero failed requests, and the average response time remained within an acceptable range. We also observed that when CPU utilization exceeded our set 70% threshold, the CloudWatch Alarm was triggered, and the Auto Scaling Group automatically increased from 1 instance to 2 instances. After the load was distributed, the system continued to operate stably, and when traffic decreased, it automatically shrunk back to its original capacity, proving that this architecture truly has the ability to dynamically scale up and down, and is not just a set-up system.

```
ab -n 50000 -c 50 http://splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com/index.php
```

```
Benchmarking splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com (be patient)
Completed 5000 requests
Completed 10000 requests
Completed 15000 requests
Completed 20000 requests
Completed 25000 requests
Completed 30000 requests
Completed 35000 requests
Completed 40000 requests
Completed 45000 requests
Completed 50000 requests
Finished 50000 requests
```

Before stress test



Key Features - High Performing

TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4



Alarms (3)

Filter alarms

Hide Auto Scaling alarms

< 1 >

EC2-High-CPU-Alarm

Metric alarm

OK

TargetTracking-splashpoint-asg-AlarmLow-1615c1fd-8617-4ffc-8b3c-4ce9ae228245

Metric alarm

In alarm

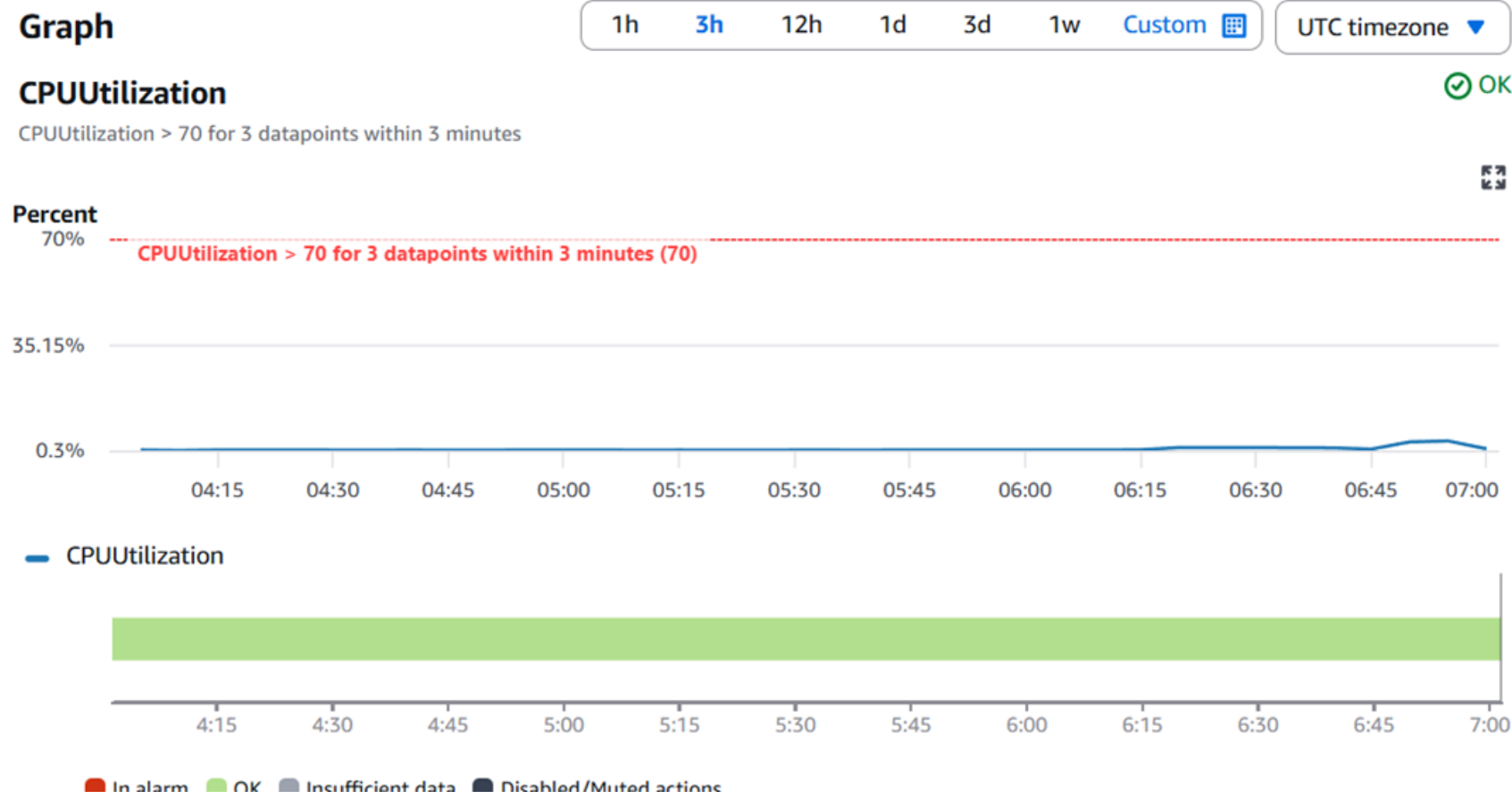
TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4

Actions

View

Explore related

Investigate



splashpoint-asg

splashpoint-asg Capacity overview

Edit

arn:aws:autoscaling:us-east-1:082810551484:autoScalingGroup:5984cd5a-bec2-4051-8294-f5129d53816f:autoScalingGroupName/splashpoint-asg

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 4	Units (number of instances)	🟢 At desired capacity

Date created
Sun Sep 06 2026 20:43:48 GMT+0800 (Malaysia Time)

- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Instances (1)

🔄 Actions ▾

🔍 Filter instances

< 1 > ⚙

☐	Instance ID ▲	Lifecycle ▼	Instance... ▼	Weighte... ▼	Launch ... ▼	Availabi... ▼	Health s... ▼	Protect... ▼
☐	i-0901cf0a66555f9da 🔗	InService	t3.micro	-	splashpoint-lt 🔗	use1-az1 (...)	🟢 Healthy	



High Performing

Key Features - Scaling Under Load

After Stress Test

```
stress: info: [12668] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [12668] successful run completed in 300s
sh-5.2$ sudo stress --cpu 2 --timeout 300s
stress: info: [13045] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [13045] successful run completed in 300s
sh-5.2$
```

```
Benchmarking splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com (be patient)
Completed 5000 requests
Completed 10000 requests
Completed 15000 requests
Completed 20000 requests
Completed 25000 requests
Completed 30000 requests
Completed 35000 requests
Completed 40000 requests
Completed 45000 requests
Completed 50000 requests
Finished 50000 requests
```

```
Server Software:      Apache/2.4.68
Server Hostname:      splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com
Server Port:          80
```

```
Document Path:        /index.php
Document Length:       14827 bytes
```

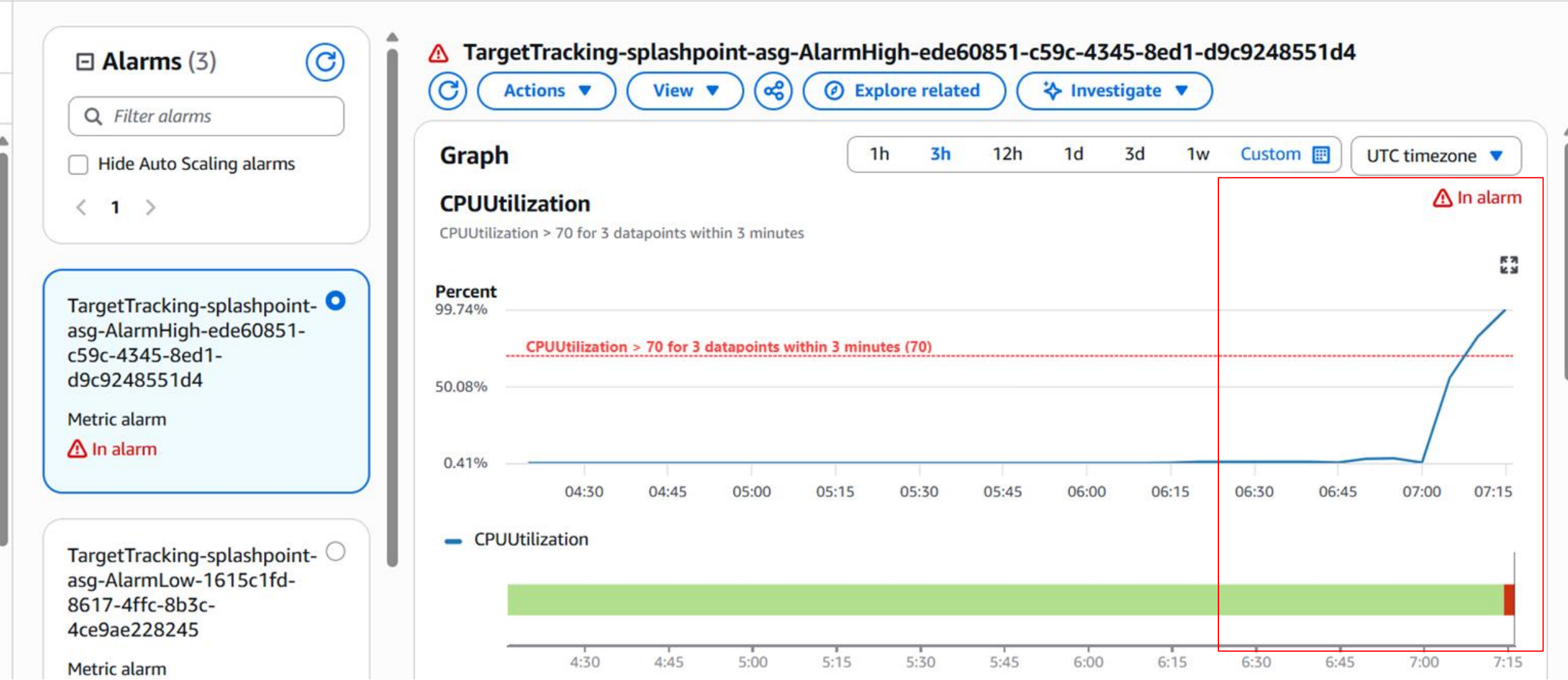
```
Concurrency Level:     50
Time taken for tests:   189.688 seconds
Complete requests:     50000
Failed requests:        0
Total transferred:      758950000 bytes
HTML transferred:       741350000 bytes
Requests per second:    263.59 [#/sec] (mean)
Time per request:       189.688 [ms] (mean)
Time per request:       3.794 [ms] (mean, across all concurrent requests)
Transfer rate:          3907.27 [Kbytes/sec] received
```

```
Connection Times (ms)
              min  mean[+/-sd] median  max
Connect:        1    2   0.6      2    31
Processing:     41   188  40.2     180   1126
Waiting:        37   173  36.2     166   1111
Total:          43   190  40.3     182   1127
```

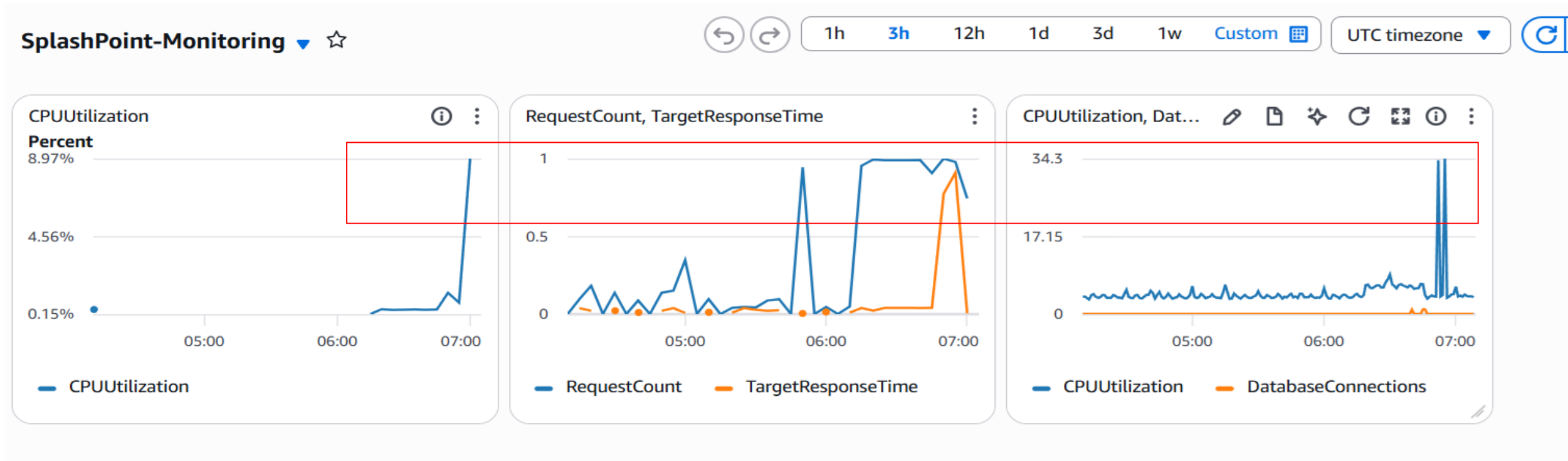
```
Percentage of the requests served within a certain time (ms)
 50%    182
 66%    191
 75%    198
 80%    203
 90%    220
 95%    253
 98%    319
 99%    358
100%   1127 (longest request)
```

Key Features - High Performing

TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4



Key Features - High Performing



Key Features - High Performing

splashpoint-asg



- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Instances (2)



Actions

Filter instances

< 1 > ⚙

☐	Instance ID	Lifecycle	Instance...	Weighte...	Launch ...	Availabi...	Health s...	Protect...
☐	i-062f00fb89ebf8218	InService	t3.micro	-	splashpoint-lt	use1-az1 (...)	Healthy	
☐	i-0975b3bdbd6318186	InService	t3.micro	-	splashpoint-lt	use1-az6 (...)	Healthy	

Status	Description	Cause	Start time	End time
Successful	Launching a new EC2 instance: i-0975b3bdbd6318186	At 2026-09-07T07:14:23Z a monitor alarm TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4 in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 1 to 2. At 2026-09-07T07:14:31Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2026 September 07, 03:14:33 PM +08:00	2026 September 07, 03:14:33 PM +08:00
	Launching a new EC2		2026	2026

Feedback



Search



^ 中 拼 网络 音量 电池

3:19 PM
9/7/2026

Key Features - High Performing

2
Total targets

✔ 2
Healthy

✖ 0
Unhealthy

0 Anomalous

⋮ 0
Unused

🕒 0
Initial

⊖ 0
Draining

▶ Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (2) Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 > ⚙

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri..
☐	i-0975b3bdbd6318186	-	80	us-east-1a (us...	✔ Healthy	-	⊖ No override.	No over
☐	i-062f00fb89ebf8218	-	80	us-east-1b (us...	✔ Healthy	-	⊖ No override.	No over



Scalability

Key Features



Scalability

Scalability — Configuration



Scalability

Key Features - Auto Scaling Group

Key Features - Scalability

☰

EC2 > Launch templates > Create launch template

ⓘ

Add new tag

You can add up to 50 more tags.

▼ Advanced details [Info](#)

IAM instance profile [Info](#)

LabInstanceProfile
arn:aws:iam::082810551484:instance-profile/LabInstanceProfile

▼

↻

Create new IAM profile

Hostname type [Info](#)

Don't include in launch template

▼

DNS Hostname [Info](#)

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

▼ Summary

Software Image (AMI)

splashpoint-ami
ami-023cbb9e0d00d0866

Virtual server type (instance type)

t3.micro

Firewall (security group)

splashpoint-sg-ec2

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

86

Key Features - Scalability

EC2 > Launch templates > Create launch template

Allow tags in metadata [Info](#)

Don't include in launch template

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
sudo dnf update -y
sudo dnf install -y httpd php php-mysql php-mbstring php-gd amazon-efs-utils
mariadb105 unzip
sudo systemctl enable httpd
sudo systemctl start httpd

curl -s https://raw.githubusercontent.com/CJY1204/Swimming-pool-system-code/main/deploy.sh | bash

DB_HOST=$(aws ssm get-parameter --name "/pool-booking/db-host" --region us-east-1 -
-query "Parameter.Value" --output text)
DB_USER=$(aws ssm get-parameter --name "/pool-booking/db-user" --region us-east-1 --
query "Parameter.Value" --output text)
DB_PASS=$(aws ssm get-parameter --name "/pool-booking/db-pass" --with-decryption --
```

▼ Summary

Software Image (AMI)

splashpoint-ami
ami-023cbb9e0d00d0866

Virtual server type (instance type)

t3.micro

Firewall (security group)

splashpoint-sg-ec2

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#)

[Create launch template](#)

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
DB_NAME=$(aws ssm get-parameter --name "/pool-booking/db-name" --region us-east-1
--query "Parameter.Value" --output text)

sudo tee /etc/httpd/conf.d/db-env.conf << EOF
SetEnv DB_HOST $DB_HOST
SetEnv DB_USER $DB_USER
SetEnv DB_PASS $DB_PASS
SetEnv DB_NAME $DB_NAME
SetEnv IMAGE_BASE_URL https://jy-pool-booking-bucket.s3.us-east-1.amazonaws.com/
EOF

systemctl restart httpd
```

☐ User data has already been base64 encoded

splashpoint-lt



splashpoint-lt (lt-0447001d2ad120248)

Actions ▼ Delete template

Launch template details

Launch template ID lt-0447001d2ad120248	Launch template name splashpoint-lt	Default version 1	Owner arn:aws:sts::082810551484:assumed-role/voclabs/user5166388=chengjy-wm25@student.tarc.edu.my
--	--	----------------------	--

Details Versions Template tags

Launch template version details

Actions ▼ Delete template version

Version 1 (Default) ▼	Description splashpoint-lt	Date created 2026-09-06T12:40:43.000Z	Created by arn:aws:sts::082810551484:assumed-role/voclabs/user5166388=chengjy-wm25@student.tarc.edu.my
--------------------------	-------------------------------	--	---

Instance details Storage Resource tags Network interfaces Advanced details

AMI ID ami-023cbb9e0d00d0866	Instance type t3.micro	Availability Zone -	Availability Zone Id -
Key pair name vockey	Security groups -	Security group IDs sg-02ed0eb567b6aa3aa	

splashpoint-asg

splashpoint-asg Capacity overview

Edit

 `arn:aws:autoscaling:us-east-1:082810551484:autoScalingGroup:5984cd5a-bec2-4051-8294-f5129d53816f:autoScalingGroupName/splashpoint-asg`

Desired capacity
2

Scaling limits
2 - 4

Desired capacity type
Units (number of instances)

Status
 Updating capacity

Date created
Sun Sep 06 2026 20:43:48 GMT+0800 (Malaysia Time)

- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Launch template

Edit

Launch template

 `lt-0447001d2ad120248`
splashpoint-lt

AMI ID

 `ami-023cbb9e0d00d0866`

Instance type
t3.micro

Owner
`arn:aws:sts::082810551484:assumed-role/voclabs/user5166388=chengjy-wm25@student.tarc.edu.my`

Version
Default

Security groups
-

Security group IDs
 `sg-02ed0eb567b6aa3aa`

Create time
Sun Sep 06 2026 20:40:43 GMT+0800 (Malaysia Time)

Description
splashpoint-lt

Storage (volumes)
-

Key pair name
vockey

Request Spot Instances
No

[View details in the launch template console](#)

splashpoint-asg

Details

Integrations

Automatic scaling

Instance management

Instance refresh

 Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, a threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when EC2 instances require more time to initialize.

Dynamic scaling policies (1) [Info](#)

Target Tracking Policy



Policy type

Target tracking scaling

Enabled or disabled

Enabled

Execute policy when

As required to maintain Average CPU utilization at 70

Take the action

Add or remove capacity units as required

Instances need

300 seconds to warm up before including in metric

Scale in

Enabled



Scalability

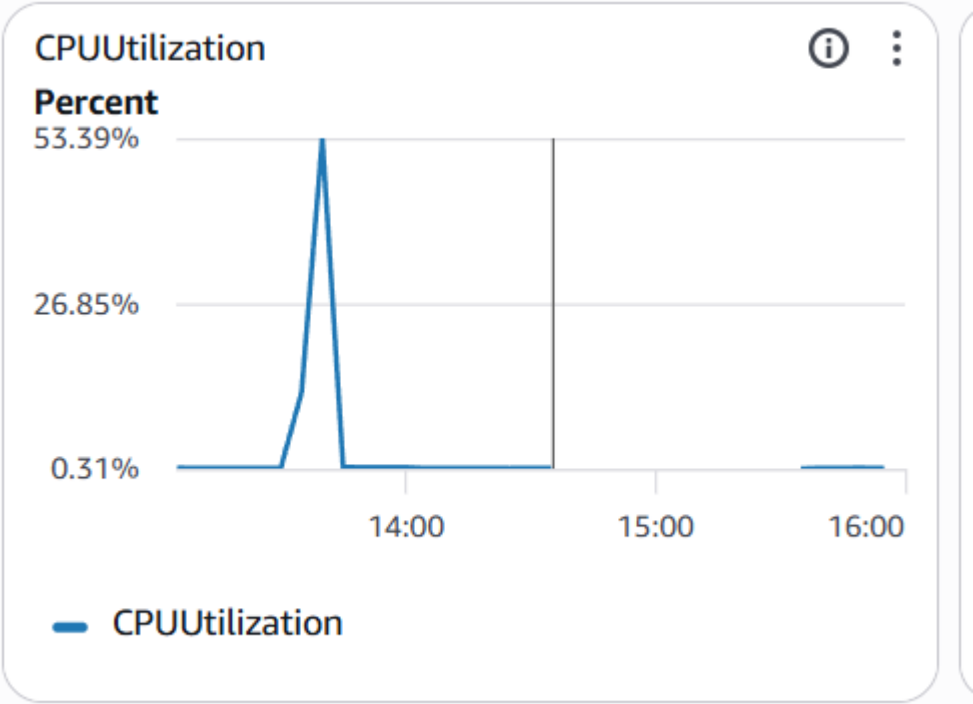
Scalability — Scale-Out & Scale-In in Action



Scalability

Scalability — Scale-Out in Action

Before stress test



Instances (2) [Info](#)

Last updated 1 minute ago

[Connect](#)

[Instance state](#) ▼

[Actions](#) ▼

[Launch instances](#) ▼

Saved filter sets

Choose filter set ▼

Filter Instance by attribute or tag (case-sensitive)

< 1 >

☐	Name	▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼	F
☐			i-062f00fb89ebf8218	✓ Running	t3.micro	✓ 3/3 checks passed	us-east-1b	-	-
☐	splashpoint-ec2		i-07b7e4582f08a0906	✓ Running	t3.micro	✓ 3/3 checks passed	us-east-1a	-	-

Key Features - Scalability

splashpoint-asg



- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Activity notifications (0)



Actions ▼

Create notification

🔍 Filter notifications

< 1 > ⚙️

☐ | Send to ▼ | On instance action ▼

No notifications are currently specified

Create notification

splashpoint-asg

splashpoint-asg Capacity overview

Edit

 arn:aws:autoscaling:us-east-1:082810551484:autoScalingGroup:5984cd5a-bec2-4051-8294-f5129d53816f:autoScalingGroupName/splashpoint-asg

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 4	Units (number of instances)	🟢 At desired capacity

Date created
Sun Sep 06 2026 20:43:48 GMT+0800 (Malaysia Time)

- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Instances (1)



Actions ▼

🔍 Filter instances

< 1 > ⚙

☐	Instance ID	▲	Lifecycle ▼	Instance... ▼	Weighte... ▼	Launch ... ▼	Availabi... ▼	Health s... ▼	Protect... ▼
☐	i-0901cf0a66555f9da 		InService	t3.micro	-	splashpoint-lt 	use1-az1 (...)	🟢 Healthy	

After stress test

TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4

Alarms (3)

Filter alarms

☐ Hide Auto Scaling alarms

< 1 >

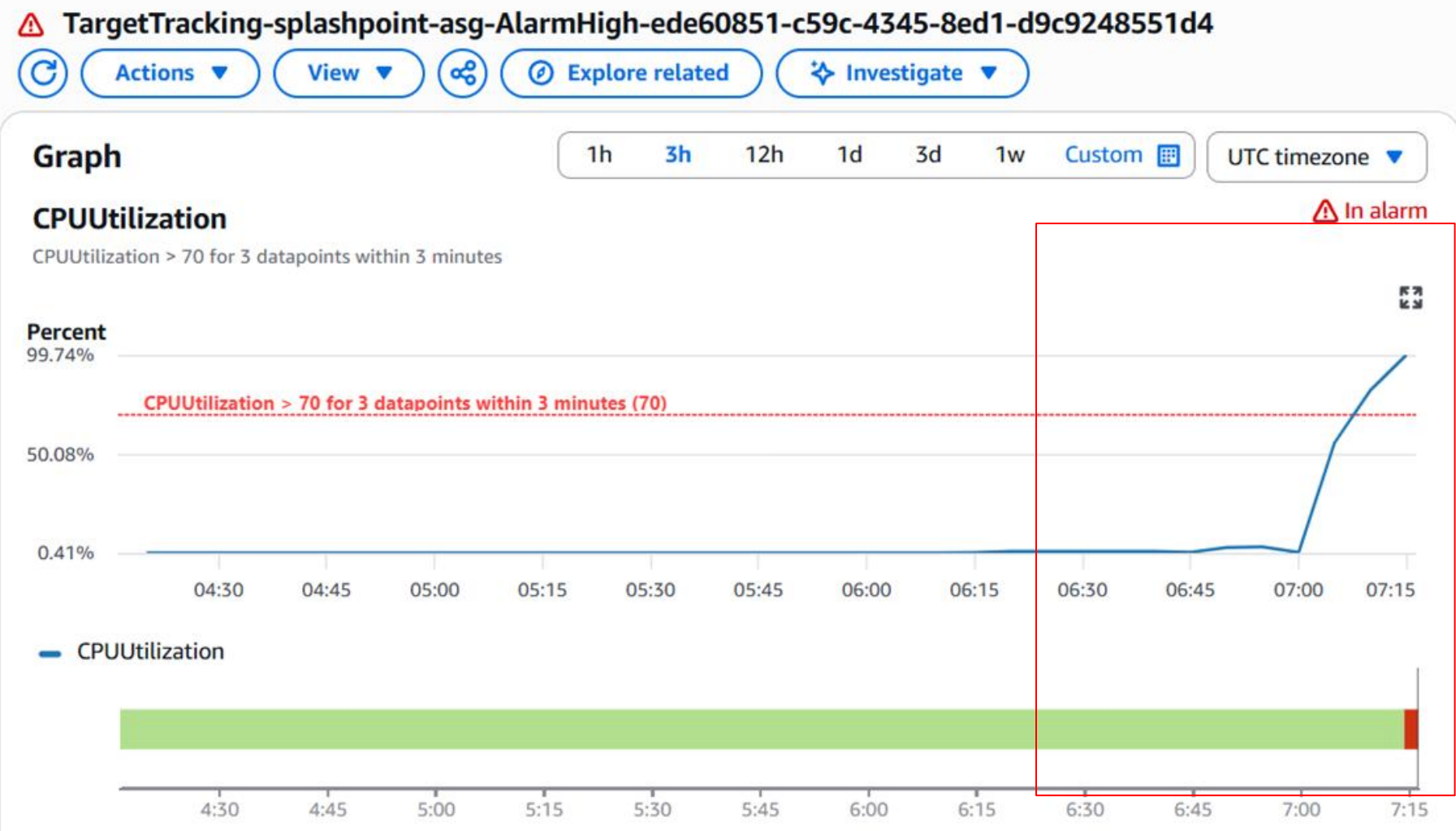
TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4

Metric alarm

In alarm

TargetTracking-splashpoint-asg-AlarmLow-1615c1fd-8617-4ffc-8b3c-4ce9ae228245

Metric alarm



Key Features - Scalability

Activity history (10)

Filter activity history

Filter by a date and time range

< 1 >

Status	Description	Cause	Start time	
Successful	Launching a new EC2 instance: i-0975b3bdbd6318186	At 2026-09-07T07:14:23Z a monitor alarm TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4 in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 1 to 2. At 2026-09-07T07:14:31Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2026 September 07, 03:14:33 PM +08:00	
	Launching a new EC2		2026	

Feedback

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Search

3:19 PM 9/7/2026

Key Features - Scalability

splashpoint-asg



- Details
- Integrations
- Automatic scaling
- Instance management
- Instance refresh
- Activity
- Monitoring
- Tags

Instances (2)



Filter instances



☐	Instance ID	Lifecycle	Instance...	Weighte...	Launch ...	Availabi...	Health s...	Protect...
☐	i-062f00fb89ebf8218	InService	t3.micro	-	splashpoint-lt	use1-az1 (...)	Healthy	
☐	i-0975b3bdbd6318186	InService	t3.micro	-	splashpoint-lt	use1-az6 (...)	Healthy	

Instances (3) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets
Choose filter set

Filter Instance by attribute or tag (case-sensitive)



☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☐		i-062f00fb89ebf8218	Running	t3.micro	3/3 checks passec	us-east-1b	-	-
☐	splashpoint-ec2	i-07b7e4582f08a0906	Running	t3.micro	3/3 checks passec	us-east-1a	-	-
☐		i-0975b3bdbd6318186	Running	t3.micro	3/3 checks passec	us-east-1a	-	-

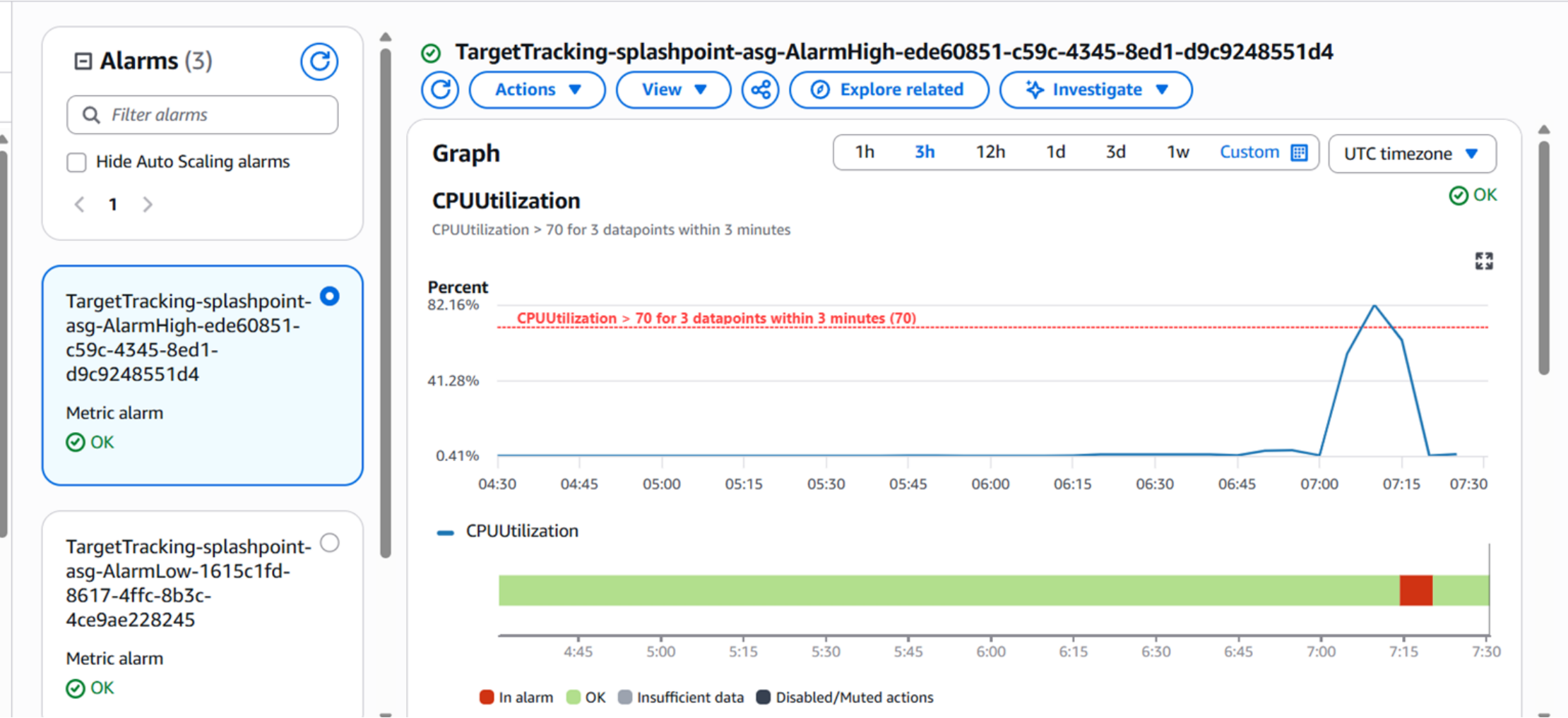


Scalability

Scalability — Scale-In in Action

Scale-In

TargetTracking-splashpoint-asg-AlarmHigh-ed60851-c59c-4345-8ed1-d9c9248551d4



Key Features - Scalability

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Instances (2)

Filter instances

Actions

< 1 >

⚙

☐	Instance ID ▾	Lifecycle ▲	Instance... ▾	Weighte... ▾	Launch ... ▾	Availabi... ▾	Health s... ▾	Protect... ▾
☐	i-062f00fb89ebf8218 ↗	InService	t3.micro	-	splashpoint-lt ↗	use1-az1 (...)	✔ Healthy	
☐	i-0975b3bdbd6318186 ↗	Terminating	t3.micro	-	splashpoint-lt ↗	use1-az6 (...)	✔ Healthy	

Instances (3) Info

Last updated 1 minute ago

Connect ↗

Instance state ▾

Actions ▾

Launch instances ▾

Saved filter sets

Choose filter set ▾

Filter Instance by attribute or tag (case-sensitive)

< 1 >

⚙

☐	Name ✎ ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Availability Zone ▾	Public IPv4 DNS ▾	F
☐		i-062f00fb89ebf8218	✔ Running ⚙ ⚙	t3.micro	✔ 3/3 checks passec	us-east-1b	-	-
☐	splashpoint-ec2	i-07b7e4582f08a0906	✔ Running ⚙ ⚙	t3.micro	✔ 3/3 checks passec	us-east-1a	-	-
☐		i-0975b3bdbd6318186	⊖ Terminated ⚙ ⚙	t3.micro	-	us-east-1a	-	-

Key Features - Scalability

splashpoint-tg

Target type

IPv4

splashpoint-alb

1

Total targets

1 Healthy

0 Unhealthy

0 Unused

0 Initial

0 Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...
☐	i-062f00fb89ebf8218	-	80	us-east-1b (us...	Healthy	-	No override.	No over

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Instances (1)



Actions



Filter instances

☐	Instance ID	Lifecycle	Instance...	Weighthe...	Launch ...	Availabi...	Health s...	Protect...
☐	i-062f00fb89ebf8218	InService	t3.micro	-	splashpoint-lt	use1-az1 (...)	Healthy	

We use a Target Tracking scaling policy, with the overall average CPU utilization of ASG as the metric, and a threshold set at 70%. When the average load exceeds this threshold, AWS will automatically calculate how many instances need to be added to bring the CPU back to near the target value, without requiring us to manually intervene to determine how many to add.



Load Balanced

Key Features



Load Balanced

Load Balanced — Configuration



Load Balanced

Key Features - Application Load Balanced

vpc-0fe59c5f43ae9ef53



vpc-0fe59c5f43ae9ef53 / splashpoint-vpc

Actions ▼

Details Info

VPC ID

vpc-0fe59c5f43ae9ef53

DNS resolution
Enabled

Main network ACL
acl-03fc0c45504e0ffd9

IPv6 CIDR (Network border group)

Encryption control ID

State

Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode

Block Public Access

⊖ Off

DHCP option set
dopt-02533e501b5c33652

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

⊗ Failed to load rule groups

DNS hostnames

Enabled

Main route table
rtb-051e60f54b8bde450

IPv6 pool

Owner ID
082810551484

Resource map

CIDRs

Flow logs

Tags

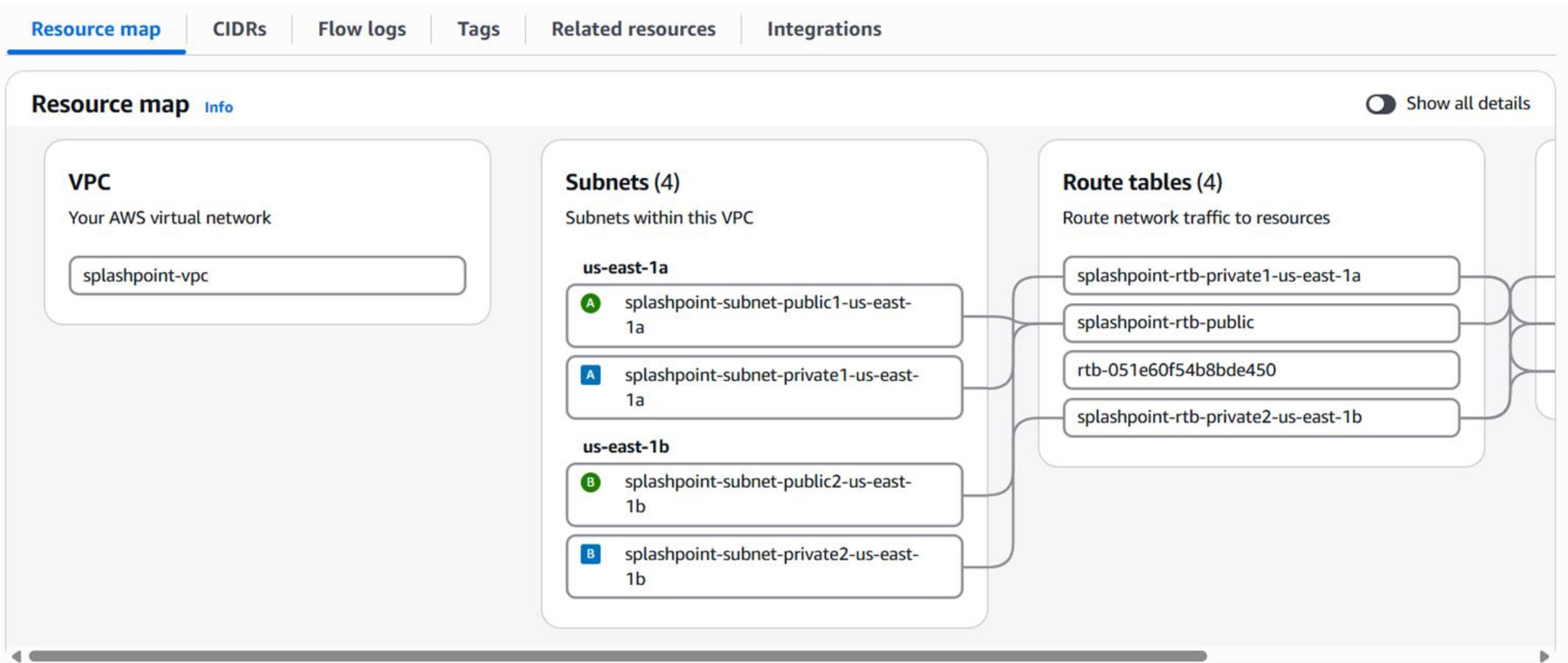
Related resources

Integrations

Resource map [Info](#)

☐ Show all details

2 AZ - 2 Public & 2 Private Subnet



splashpoint-alb



splashpoint-alb



Actions ▼

▼ Details

Load balancer type
Application

Scheme
Internet-facing

Status
✔ Active

Hosted zone
Z35SXDOTRQ7X7K

VPC
[vpc-0fe59c5f43ae9ef53](#)

Availability Zones
[subnet-0f27dccf81d9722c8](#) us-east-1b (use1-az1)
[subnet-026297abbd88bd23a](#) us-east-1a (use1-az6)

Load balancer IP address type
IPv4

Date created
September 6, 2026, 20:42 (UTC+08:00)

Load balancer ARN
 `arn:aws:elasticloadbalancing:us-east-1:082810551484:loadbalancer/app/splashpoint-alb/514640b9dd1f9348`

DNS name [Info](#)
 `splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com` (A Record)

Key Features - Load Balanced

2
Total targets

✔ 2
Healthy
0 Anomalous

✖ 0
Unhealthy

⋮ 0
Unused

⌚ 0
Initial

⊖ 0
Draining

► **Distribution of targets by Availability Zone (AZ)**

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2) Info

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

< 1 > ⚙

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri..
☐	i-0975b3bdbd6318186	-	80	us-east-1a (us...	✔ Healthy	-	⊖ No override.	No over
☐	i-062f00fb89ebf8218	-	80	us-east-1b (us...	✔ Healthy	-	⊖ No override.	No over

Our ALB spans two Availability Zones, and instances under the Target Group are also distributed across these two Availability Zones. This design aims to ensure that even if one Availability Zone experiences a problem, instances in the other Availability Zone can continue to provide service, preventing a single point of failure.



Load Balanced

Load Balanced — Even Traffic Distribution

Using : `sudo wc -l /var/log/httpd/access_log`
Before sending request

```
sh-5.2$ sudo wc -l /var/log/httpd/access_log
53395 /var/log/httpd/access_log
sh-5.2$
```

```
sh-5.2$ sudo wc -l /var/log/httpd/access_log
1438 /var/log/httpd/access_log
sh-5.2$
```

After sending request using :

ab -n 5000 -c 200 http://splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com/index.php

```
sh-5.2$ sudo wc -l /var/log/httpd/access_log
53395 /var/log/httpd/access_log
sh-5.2$ sudo wc -l /var/log/httpd/access_log
55903 /var/log/httpd/access_log
sh-5.2$
```

$$55903 - 53395 = 2508$$

```
sh-5.2$ sudo wc -l /var/log/httpd/access_log
1438 /var/log/httpd/access_log
sh-5.2$ sudo wc -l /var/log/httpd/access_log
3866 /var/log/httpd/access_log
sh-5.2$
```

$$3866 - 1438 = 2428$$

two instances differed by at least less than [5%]

By comparing the number of access log requests for each instance, we verified that the ALB traffic distribution was indeed even – the number of requests handled by the two instances differed by less than [5%], proving that no single instance was overloaded with requests and that the load was indeed evenly distributed, rather than being carried by a single instance.



Additional Services



Additional Services - Lambda + SNS

SNS Topic

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

pool-booking-confirmations

pool-booking-confirmations

EditDeletePublish message

Details

Name

pool-booking-confirmations

ARN

arn:aws:sns:us-east-1:082810551484:pool-booking-confirmations

Display name

-

Type

Standard

Topic owner

082810551484

Subscriptions

Access policy

Delivery policy (HTTP/S)

Delivery status logging

Encryption

Tags

Integrations

Subscriptions (1)

EditDeleteRequest confirmationConfirm subscriptionCreate subscription

Search

< 1 > ⚙

ID	Endpoint	Status	Protocol
☐ 0935e91e-10a1-40ab-8389-3c90af...	arn:aws:lambda:us-east-1:0828105...	✔ Confirmed	LAMBDA

118

Create lambda & it function

The screenshot displays the AWS Lambda console interface for a function named 'send-booking-confirmation'. The left sidebar shows the 'EXPLORER' view with the function's code file 'lambda_function.py' selected. The main editor area shows the Python code for the lambda function, which imports 'json', 'smtplib', and 'MIMEText' from 'email.mime.text'. It defines a 'lambda_handler' function that processes an event record, extracts booking details (recipient, name, pool, date, time, reference), and sends an email confirmation. The code includes a 'body' string with placeholders for the booking details and a 'msg' object for the email content. The function returns a status code of 200.

```
1 import json
2 import smtplib
3 from email.mime.text import MIMEText
4
5 GMAIL_ADDRESS = "project.use0421@gmail.com"
6 GMAIL_APP_PASSWORD = "gamlzflueoswwhii"
7
8 def lambda_handler(event, context):
9     for record in event['Records']:
10         message = json.loads(record['Sns']['Message'])
11
12         recipient = message.get('email')
13         name = message.get('name')
14         pool = message.get('pool')
15         date = message.get('date')
16         time = message.get('time')
17         reference = message.get('reference')
18
19         body = f"""Hi {name},
20
21 Your booking is confirmed!
22 {pool}
23 Date: {date}
24 Time: {time}
25 """
26
27         msg = MIMEText(body)
28         msg['Subject'] = f"Booking Confirmed - {reference}"
29         msg['From'] = GMAIL_ADDRESS
30         msg['To'] = recipient
31
32         with smtplib.SMTP('smtp.gmail.com', 587) as server:
33             server.starttls()
34             server.login(GMAIL_ADDRESS, GMAIL_APP_PASSWORD)
35             server.sendmail(GMAIL_ADDRESS, [recipient], msg.as_string())
36
37         print(f"Email sent to {recipient} for booking {reference}")
38
39     return {'statusCode': 200}
```

Create Subscriptions

≡ [Amazon SNS](#) > [Subscriptions](#) > Create subscription

Create subscription

Details

Topic ARN

arn:aws:sns:us-east-1:082810551484:pool-booking-confirmations



Protocol

The type of endpoint to subscribe

AWS Lambda



Endpoint

An AWS Lambda function that can receive notifications from Amazon SNS.

arn:aws:lambda:us-east-1:082810551484:function:send-booking-confirmation



Lambda > Functions > send-booking-confirmation

Diagram | Template

send-booking-confirmation

Layers (0)

SNS

+ Add trigger

+ Add destination

Function ARN
 arn:aws:lambda:us-east-1:082810551484:function:send-booking-confirmation

Description
-

Last modified
2 weeks ago

Code | Test | Monitor

Configuration Aliases Versions

Triggers (1) Info

Search for triggers

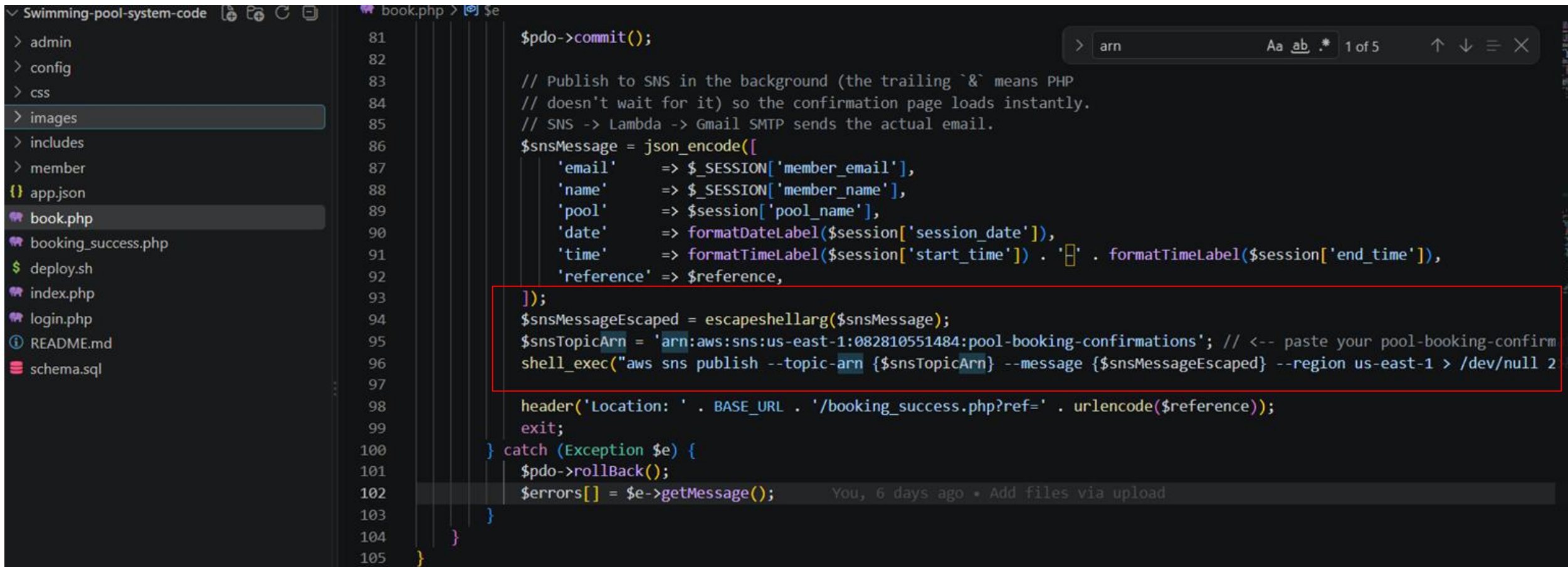
☐ Trigger

SNS: pool-booking-confirmations
arn:aws:sns:us-east-1:082810551484:pool-booking-confirmations
► Details

Fix errors Edit Delete Add trigger

< 1 >

Inside the code use back the SNS arn link to connect



```
Swimming-pool-system-code
> admin
> config
> css
> images
> includes
> member
{} app.json
book.php
booking_success.php
$ deploy.sh
index.php
login.php
README.md
schema.sql

81 $pdo->commit();
82
83 // Publish to SNS in the background (the trailing `&` means PHP
84 // doesn't wait for it) so the confirmation page loads instantly.
85 // SNS -> Lambda -> Gmail SMTP sends the actual email.
86 $snsMessage = json_encode([
87     'email' => $_SESSION['member_email'],
88     'name' => $_SESSION['member_name'],
89     'pool' => $session['pool_name'],
90     'date' => formatDateLabel($session['session_date']),
91     'time' => formatTimeLabel($session['start_time']) . ' - ' . formatTimeLabel($session['end_time']),
92     'reference' => $reference,
93 ]);
94 $snsMessageEscaped = escapeshellarg($snsMessage);
95 $snsTopicArn = 'arn:aws:sns:us-east-1:082810551484:pool-booking-confirmations'; // <-- paste your pool-booking-confirm
96 shell_exec("aws sns publish --topic-arn {$snsTopicArn} --message {$snsMessageEscaped} --region us-east-1 > /dev/null 2
97
98 header('Location: ' . BASE_URL . '/booking_success.php?ref=' . urlencode($reference));
99 exit;
100 } catch (Exception $e) {
101     $pdo->rollBack();
102     $errors[] = $e->getMessage();
103 }
104 }
105 }
```

Book: The Deep Blue

Tomorrow · 7:00 AM–12:00 PM | 40 spots left | Free

Booking as JUN YU CHENG (chengjy-wm25@student.tarc.edu.my)

Tickets 1 (one booking per person)

Cost Free 🎉

Confirm Booking

Cancel



Booking Confirmed!

Thank you, JUN YU CHENG. Your reservation is set.

SPB - A525121A

The Deep Blue

Tomorrow

7:00 AM – 12:00 PM

Tickets: 1 | Cost: Free

A confirmation has been recorded under chengjy-wm25@student.tarc.edu.my. You can find this booking any time in your profile.

Book Another Session

View My Bookings

The screenshot displays the Gmail web interface. On the left is a sidebar with navigation options: Mail (with a 99+ badge), Chat, and Meet. The main area on the left shows the 'Compose' button and a list of folders: 'Inbox' (987), 'Starred', 'Snoozed', 'Sent', 'Drafts' (7), 'Categories', 'Purchases' (4), and 'More'. At the bottom of this list is a 'Labels' section with a plus icon. The top of the main area features a search bar labeled 'Search mail' and a settings icon. The email being viewed is titled 'Booking Confirmed - SPB-A525121A' and is marked as 'External' and 'Inbox'. The sender is 'project.use0421@gmail.com' and the recipient is 'me'. The email body contains the following text: 'Hi JUN YU CHENG, Your booking is confirmed! Pool: The Deep Blue Date: Tomorrow Time: 7:00 AM–12:00 PM Reference: SPB-A525121A See you at the pool! SplashPoint'.

Gmail

Search mail

99+ Mail

Compose

Inbox 987

Starred

Snoozed

Sent

Drafts 7

Categories

Purchases 4

More

Labels +

Booking Confirmed - SPB-A525121A External Inbox x

project.use0421@gmail.com to me

Hi JUN YU CHENG,

Your booking is confirmed!

Pool: The Deep Blue
Date: Tomorrow
Time: 7:00 AM–12:00 PM
Reference: SPB-A525121A

See you at the pool!
SplashPoint

Key Features - Load Balanced

CloudWatch Logs

Lambda logs all requests handled by your function and automatically stores logs generated by your code through Amazon CloudWatch Logs. To validate your code, instrument it with custom logging statements. The following tables list the most recent and most expensive function invocations across all function activity. To view logs for a specific function version or alias, visit the **Monitor** section at that level.

Recent invocations

#	Timestamp	RequestId	LogStream	DurationInMS	BilledDurationInMS	MemorySetInMB	MemoryUsed
▶ 1	2026-09-12T13:20:32.388Z	3ac9906c-eee1-494d-83c9-dee411b41c48	2026/09/12/[\$LATEST]2419dab61a244cf29ed468602666aced	1228.33	1357.0	128.0	51.0

ient > /aws/lambda/send-booking-confirmation > 2026/09/12/[\$LATEST]2419dab61a244cf29ed468602666aced

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear1m30m1h12hCustomUTC timezone

Display

▶	Timestamp	Message
▶	2026-09-12T13:20:31.028Z	INIT_START Runtime Version: python:3.12.mainlinev2.v31 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:c1ab740f3656a72d79176...
▼	2026-09-12T13:20:31.159Z	START RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48 Version: \$LATEST
		START RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48 Version: \$LATEST
▼	2026-09-12T13:20:32.386Z	Email sent to chengjy-wm25@student.tarc.edu.my for booking SPB-A525121A
		Email sent to chengjy-wm25@student.tarc.edu.my for booking SPB-A525121A
▼	2026-09-12T13:20:32.388Z	END RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48
		END RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48
▼	2026-09-12T13:20:32.388Z	REPORT RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48 Duration: 1228.33 ms Billed Duration: 1357 ms Memory Size: 128 MB Max Memory...
		REPORT RequestId: 3ac9906c-eee1-494d-83c9-dee411b41c48 Duration: 1228.33 ms Billed Duration: 1357 ms Memory Size: 128 MB Max Memory Used: 51

Back to top



Additional Services - S3

S3 bucket

Amazon S3 > Buckets > jy-pool-booking-bucket

jy-pool-booking-bucket

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

☐	Name	Type	Last modified	Size	Storage class
☐	 coral-cove.jpg	jpg	September 2, 2026, 16:50:46 (UTC+08:00)	2.9 MB	Standard
☐	 deep-blue.jpg	jpg	September 2, 2026, 16:50:48 (UTC+08:00)	3.5 MB	Standard
☐	 tide-pool.jpg	jpg	September 2, 2026, 16:50:50 (UTC+08:00)	3.5 MB	Standard

127

Using SetEnv to connect s3 bucket

```
sudo tee /etc/httpd/conf.d/db-env.conf << EOF
```

```
SetEnv DB_HOST $DB_HOST
```

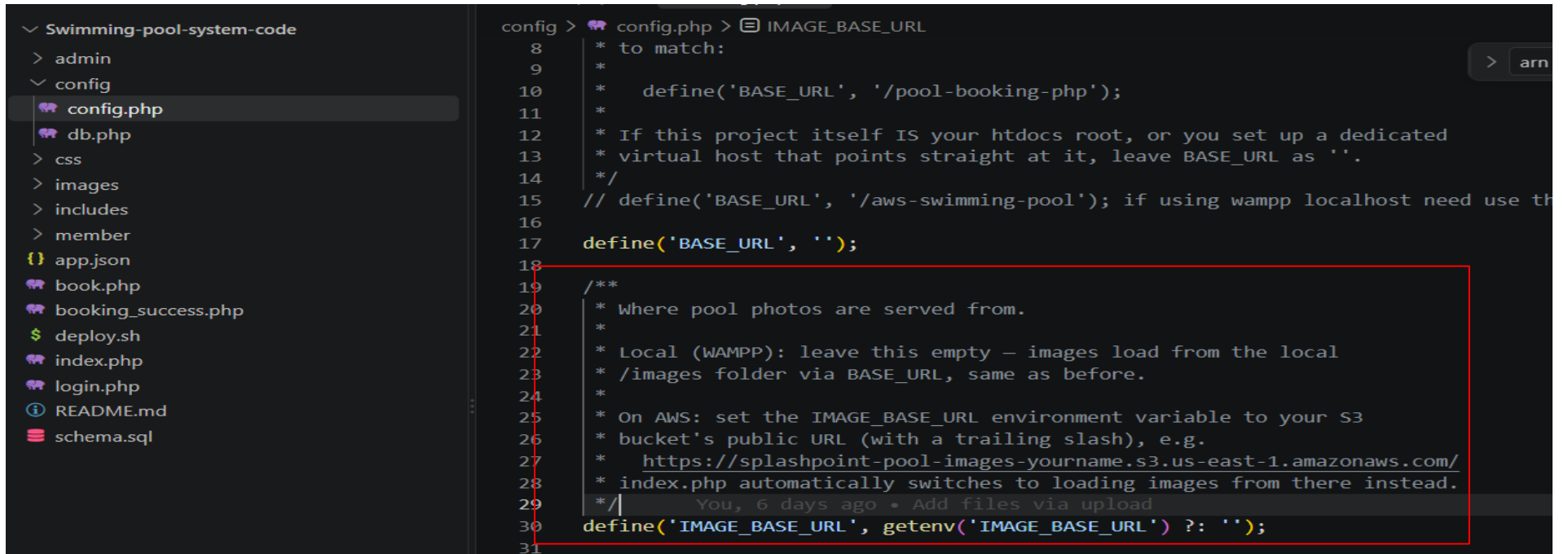
```
SetEnv DB_USER $DB_USER
```

```
SetEnv DB_PASS $DB_PASS
```

```
SetEnv DB_NAME $DB_NAME
```

```
SetEnv IMAGE_BASE_URL https://jy-pool-booking-bucket.s3.us-east-1.amazonaws.com/
```

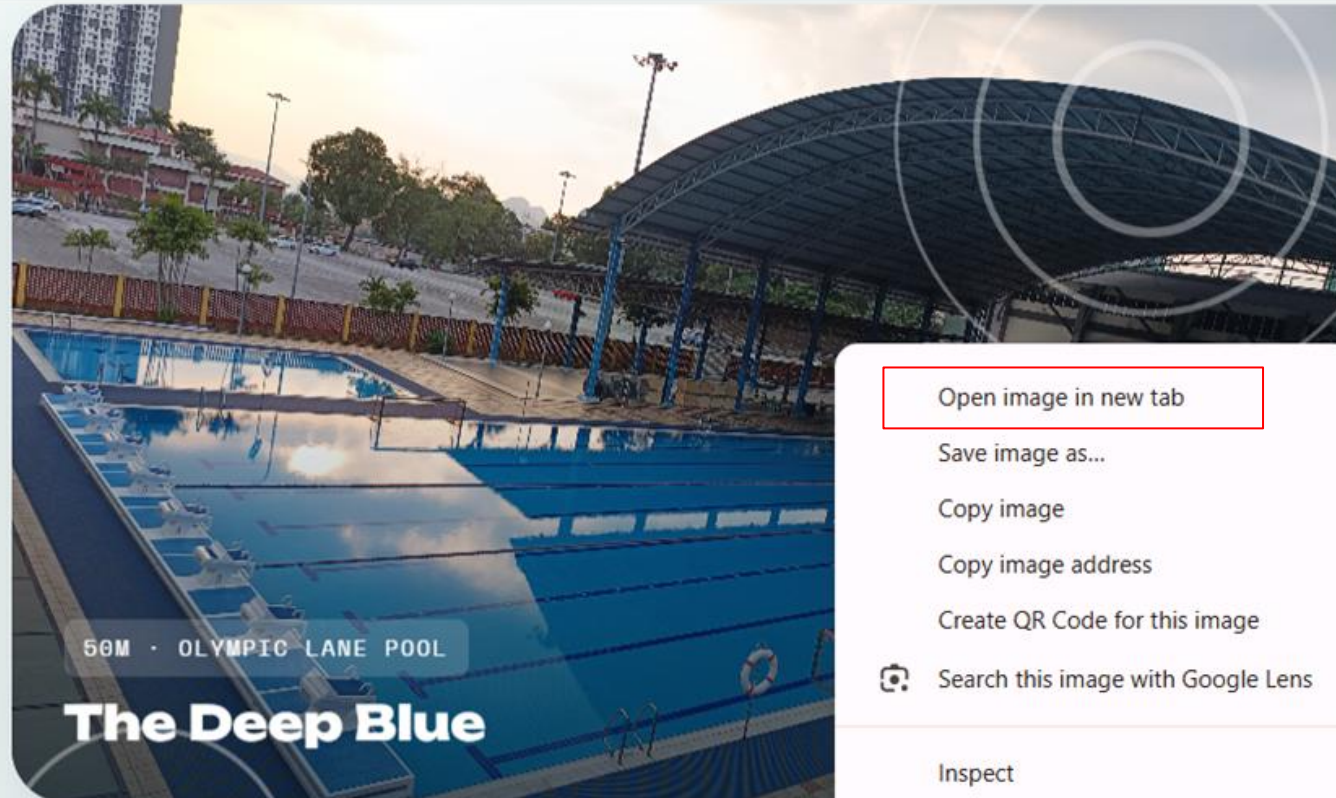
```
EOF
```



```
config > config.php > IMAGE_BASE_URL
8  *
9  * to match:
10 *
11 *   define('BASE_URL', '/pool-booking-php');
12 *
13 * If this project itself IS your htdocs root, or you set up a dedicated
14 * virtual host that points straight at it, leave BASE_URL as ''.
15 */
16 // define('BASE_URL', '/aws-swimming-pool'); if using wampp localhost need use th
17 define('BASE_URL', '');
18
19 /**
20  * Where pool photos are served from.
21  *
22  * Local (WAMPP): leave this empty – images load from the local
23  * /images folder via BASE_URL, same as before.
24  *
25  * On AWS: set the IMAGE_BASE_URL environment variable to your S3
26  * bucket's public URL (with a trailing slash), e.g.
27  * https://splashpoint-pool-images-yourname.s3.us-east-1.amazonaws.com/
28  * index.php automatically switches to loading images from there instead.
29  */
30 define('IMAGE_BASE_URL', getenv('IMAGE_BASE_URL') ?: '');
31
```

Key Features - Load Balanced

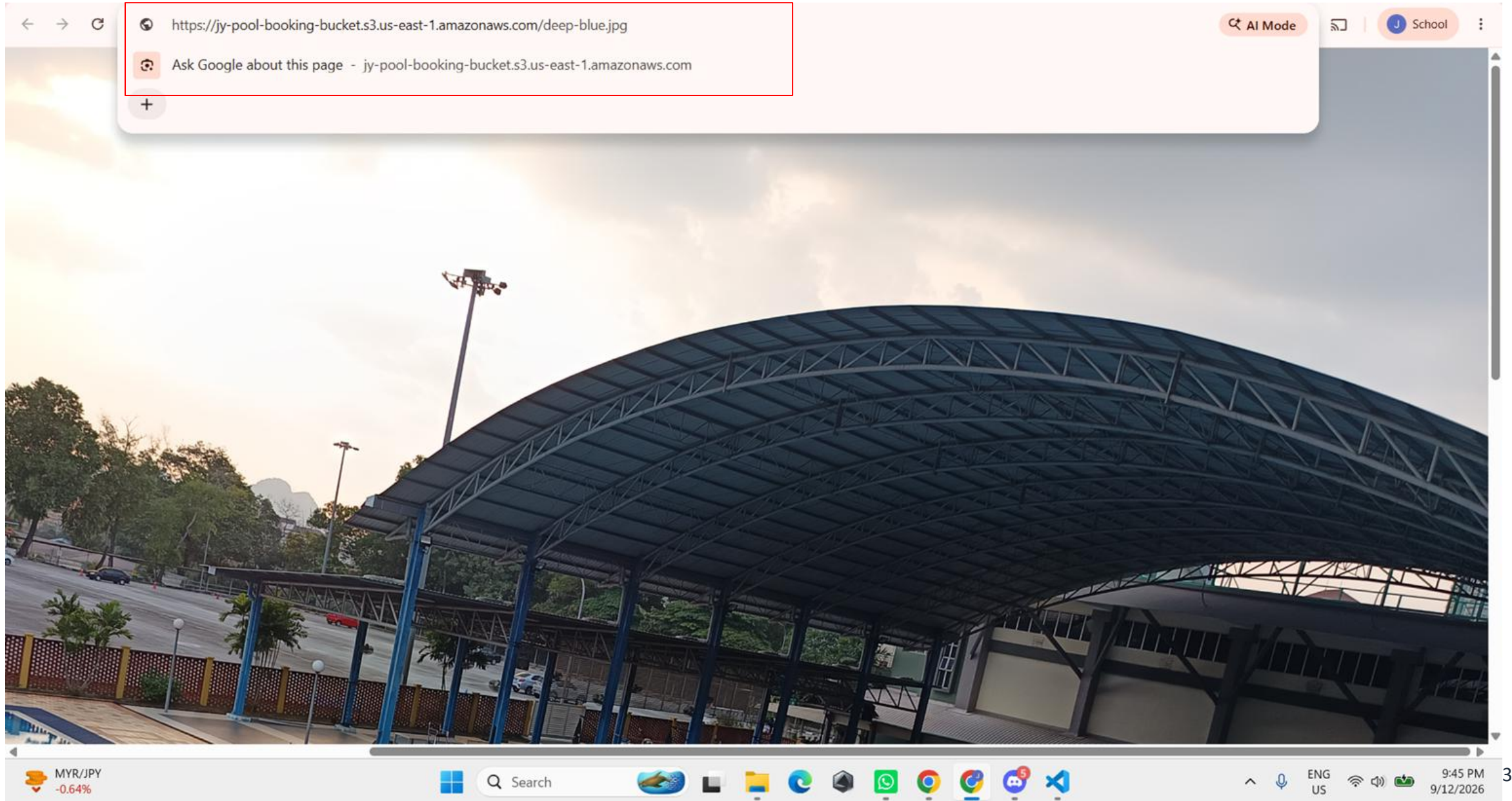
⚠ Not secure splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com/index.php

[Pools](#)[My Profile](#)[Log out](#)

Our full-length competition pool with 8 marked lanes. Ideal for serious lap swimmers, squad training, and anyone who wants real distance.

DEPTH**1.2m - 2.0m****CAPACITY****40 people****NEXT SESSION****Tomorrow 7:00 AM****3%****NEXT SESSION****39 spots left****FREE****Tomorrow · 7:00 AM-12:00 PM****39 left****Tomorrow · 2:00 PM-7:00 PM****40 left**

Key Features - Load Balanced



Key Features - Load Balanced



PoolsMy ProfileLog out



50M · OLYMPIC LANE POOL

The Deep Blue

Our full-length competition pool with 8 marked lanes. Ideal for serious lap swimmers, squad training, and anyone who wants real distance.

DEPTH

1.2m - 2.0m

CAPACITY

40 people

NEXT SESSION

Tomorrow 7:00 AM

3% NEXT SESSION

39 spots left

FREE

Tomorrow · 7:00 AM-12:00 PM

39 left

Tomorrow · 2:00 PM-7:00 PM

40 left

3 / 10 requests0.0 kB / 13.3 kB tr

Not secure splashpoint-alb-1364108331.us-east-1.elb.amazonaws.com/index.php

ElementsConsoleSourcesNetwork

Filter

deep-blue.jpg

General

Request URLhttps://jy-pool-booking-buck et.s3.us-east-1.amazonaws.co m/deep-blue.jpg

Request MethodGET

Status Code200 OK (from memory cac he)

Remote Address16.15.213.104:443

Referrer Policystrict-origin-when-cross-origi n

Response headers

Accept-Rangesbytes

Content-Length3688697

Content-Typeimage/jpeg

DateSat, 12 Sep 2026 07:14:54 GMT

Etag"d807ff0626115a627 189a04d5b7573ac"

Last-ModifiedWed, 02 Sep 2026 08:50:48 GMT

ServerAmazonS3

X-Amz-Id-2nGp4ScXT12oFTg0SMF kh4Z9LZUH13qY6PDgB



Additional Services - Cloud Trail

Key Features - Load Balanced

- Step 1
- Choose trail attributes
- Step 2
- Choose log events
- Step 3
- Review and create

Choose trail attributes

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name

Enter a display name for your trail.

pool-booking-audit-trail

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization

To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location

[Info](#)

☒ Create new S3 bucket

Create a bucket to store logs for the trail.

☐ Use existing S3 bucket

Choose an existing bucket to store logs for this trail.

Trail log bucket and folder

Trails

Copy events to Lake



Delete

Create trail



	Name ▲	Home region ▼	Multi-region trail ▼	ARN ▼	Insights ▼	Organization trail ▼	S3 bucket ▼	Log file prefix ▼	CloudWatch L log group
☐	pool-booking-audit-trail	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:082810551484:trail/pool-booking-audit-trail	Disabled	No	aws-cloudtrail-logs-082810551484-796d9ede	-	-

Key Features - Load Balanced

☰

CloudTrail > Trails > arn:aws:cloudtrail:us-east-1:082810551484:trail/pool-booking-audit-trail

ⓘ

pool-booking-audit-trail

Delete

Stop logging

General details

Edit

Trail logging

✔ Logging

Trail name

pool-booking-audit-trail

Multi-region trail

Yes

Apply trail to my organization

Not enabled

SNS notification delivery

Disabled

Last SNS notification

-

Trail log location

[aws-cloudtrail-logs-082810551484-796d9ede/AWSLogs/082810551484](#) 🔗

Last log file delivered

September 13, 2026, 12:39:22 (UTC+08:00)

Log file SSE-KMS encryption

Not enabled

Log file validation

Enabled

Last file validation delivered

September 13, 2026, 12:29:08 (UTC+08:00)

Recursive logging

Enabled

Management events

Edit

API activity

All

Exclude AWS KMS events

No

Exclude Amazon RDS Data API events

No



Conclusion

Challenges & solutions

- Learner Lab IAM restrictions blocked certain services (CloudFront, SES) — tested cheaply first, pivoted to alternatives (Gmail SMTP)
- PHP sessions broke across multiple EC2 instances behind the ALB — solved with a database-backed session handler
- New ASG instances launched with outdated configuration — solved by moving setup logic into Launch Template User Data

AWS resources used

- AWS Systems Manager (Parameter Store) documentation
- AWS WAF Managed Rule Groups
- AWS Pricing Calculator
- AWS Academy Learner Lab service-restriction notes

New skills gained

- Auto Scaling Group configuration with Launch Templates & User Data bootstrapping
- IAM least-privilege Security Group design
- Serverless event-driven architecture (SNS + Lambda)
- Load testing methodology (Apache Bench)

Future plan

- Migrate to HTTPS with a custom domain + ACM certificate
- Automate deployment via S3 + Instance Refresh (CI/CD-style)
- Move email confirmation to Amazon SES once verified for production
- Explore VPC Endpoints to remove NAT Gateway dependency



References



References - Official Cloud & Service Documentation

AWS Official Documentation:

- Amazon Web Services. (2024). *Amazon Elastic Compute Cloud (EC2) User Guide*. AWS Documentation. <https://docs.aws.amazon.com/ec2/>
- Amazon Web Services. (2024). *Application Load Balancer Documentation*. AWS Documentation. <https://docs.aws.amazon.com/alb/>
- Amazon Web Services. (2024). *Amazon Aurora & RDS User Guide*. AWS Documentation. <https://docs.aws.amazon.com/rds/>
- Amazon Web Services. (2024). *Auto Scaling Groups User Guide*. AWS Documentation. <https://docs.aws.amazon.com/autoscaling/>
- Amazon Web Services. (2024). *AWS Systems Manager Parameter Store User Guide*. AWS Documentation. <https://docs.aws.amazon.com/systems-manager/latest/userguide/systems-manager-parameter-store.html>
- Amazon Web Services. (2024). *AWS WAF, AWS Shield, and AWS Firewall Manager Developer Guide*. AWS Documentation. <https://docs.aws.amazon.com/waf/>



References - Tools, Web Technologies & Open Source Libraries

Performance Testing & Benchmarking:

- Apache Software Foundation. (2024). *ab - Apache HTTP server benchmarking tool*. Apache HTTP Server Documentation. <https://httpd.apache.org/docs/2.4/programs/ab.html>

Core Application Stack & Languages:

- PHP Documentation Group. (2024). *PHP: Hypertext Preprocessor Documentation*. PHP.net. <https://www.php.net/docs.php>
- The MySQL Team. (2024). *MySQL 8.0 Reference Manual*. Oracle Corporation. <https://dev.mysql.com/doc/refman/8.0/en/>

AWS Pricing & Estimation Tools:

- Amazon Web Services. (2024). *AWS Pricing Calculator*. <https://calculator.aws/>



Appendix (If any) - None



Thank you